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Project Name:

Flight Delay Analysis

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Abstract

This project focuses on analyzing the causes of domestic flight delays in the United States during the period from **January 1, 2019, to June 30, 2019**. The main objective is to identify and evaluate the key factors contributing to flight delays, measure their impact, and explore potential ways to minimize them.

The analysis was conducted using modern data analysis tools to examine relationships between delay causes, delay durations, and the most affected airlines and airports. By exploring these patterns, the project aims to provide a deeper understanding of operational inefficiencies within the aviation system.

The results indicate that **"Late Aircraft"** is the primary cause of delays, followed by **weather-related issues**. These findings highlight the need for better scheduling strategies, improved maintenance planning, and enhanced airport infrastructure management.

Ultimately, the insights from this analysis can support airlines and airport authorities in improving flight operations, optimizing runway usage, and reducing overall delay occurrences.

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Chapter 1

Introduction

General Introduction

Air transportation plays a vital role in connecting people, businesses, and economies around the world. In the United States, domestic air travel represents one of the most active and complex transportation networks globally, with thousands of flights operating daily across hundreds of airports.

However, flight delays remain a persistent challenge that significantly affects the efficiency and reliability of air transport. Delays not only lead to financial losses for airlines and airports but also cause inconvenience for passengers, affecting their travel schedules and overall satisfaction. Moreover, flight delays can create a chain reaction in the aviation system — a single delayed aircraft may result in multiple subsequent delays throughout the day.

Understanding the factors that contribute to these delays is essential for improving operational performance and passenger experience. Several factors can lead to flight delays, including adverse weather conditions, air traffic control restrictions, airline operational issues, security concerns, and late arrivals of aircraft from previous flights. While some causes are unavoidable, many can be mitigated through better planning, scheduling, and resource allocation.

With the rapid growth of data availability and analytical tools, data analysis has become a powerful method for investigating complex problems such as flight delays. Through careful examination of large datasets, it is possible to identify patterns, correlations, and root causes that may not be immediately visible.

This project focuses on analyzing the causes of domestic flight delays in the United States during the period from **January 1, 2019, to June 30, 2019**. Using a dataset that includes detailed information about flights and the reasons for their delays, the study aims to uncover the most influential factors affecting flight punctuality. By leveraging modern data analysis tools, the project provides insights that can help airlines and airport authorities improve decision-making, enhance operational efficiency, and reduce the overall occurrence of flight delays.

Ultimately, this analysis not only contributes to understanding the operational behavior of the aviation sector but also demonstrates the value of data analytics as a key driver for smarter, evidence-based management within the transportation industry.

Chapter 2

Background

Background

Air travel has become one of the most essential modes of transportation in the modern world, enabling the rapid movement of people and goods across vast distances. In the United States, the domestic aviation network is one of the largest and busiest in the world, with thousands of flights scheduled daily across multiple states and time zones. However, the increasing volume of air traffic has also introduced greater complexity in managing flight operations, leading to one of the most persistent challenges in the aviation industry — **flight delays**.

Flight delays occur when scheduled flights do not depart or arrive on time, and they have wide-ranging implications. From an operational perspective, delays cause significant financial losses for airlines due to fuel waste, crew overtime, and disrupted schedules. For passengers, delays lead to frustration, missed connections, and decreased satisfaction with airline services. Additionally, flight delays have a cascading effect on airport traffic and logistics, potentially disrupting the broader transportation network.

The causes of flight delays are multifaceted. Common reasons include adverse **weather conditions**, **air traffic congestion**, **aircraft maintenance or technical issues**, **security procedures**, and **late arrivals of aircraft from previous flights**. Some of these causes are beyond human control, such as weather-related factors, while others can be mitigated through effective management and planning.

In recent years, the availability of large-scale flight data and advances in **data analytics** have created new opportunities to better understand the dynamics of flight delays. By analyzing detailed datasets that include information about flight times, airlines, airports, and delay causes, researchers and industry professionals can identify underlying patterns and develop strategies to minimize operational inefficiencies.

This project utilizes a dataset covering U.S. domestic flights between **January 1, 2019, and June 30, 2019**, aiming to analyze the main causes of flight delays and evaluate their impact on overall performance. Through data-driven insights, the study seeks to support airlines and airport authorities in developing more efficient scheduling systems, optimizing airport operations, and ultimately improving the reliability of air travel in the United States.

Chapter 3

Methodology

3.1 Dataset

The dataset used in this project is titled **"Flight Delay and Causes"**, obtained from the open data platform **Kaggle** (https://www.kaggle.com/datasets/undersc0re/flight-delay-and-causes). It contains detailed records of **U.S. domestic flights** between **January 1, 2019, and June 30, 2019**, providing extensive information about flight schedules, delays, and the reasons behind them.

The dataset was compiled from the **U.S. Department of Transportation** official records and made publicly available for research and analysis purposes. It is stored in **CSV format**, which makes it easy to process and integrate with various data analysis tools such as Python (Pandas), Excel, and Power BI.

Dataset Overview

- **Number of rows:** 484,552 (each row represents one flight record)
- **Number of columns:** 28
- **File type:** CSV (Comma-Separated Values)
- **Data size:** approximately 65 MB

The dataset covers multiple U.S. airlines and airports, representing a large and diverse sample suitable for exploratory data analysis, statistical correlation, and visualization of delay causes.

Main Columns Description

- **DayOfWeek:** Indicates the day of the week the flight took place (1 = Monday, 7 = Sunday).
- **Date:** The full date of the flight (YYYY-MM-DD format).
- **DepTime:** The actual departure time of the flight (in local time).
- **ArrTime:** The actual arrival time of the flight (in local time).
- **CRSArrTime:** The scheduled arrival time according to the airline's published timetable.
- **UniqueCarrier:** The airline's code identifier (e.g., AA for American Airlines, DL for Delta Air Lines).
- **Airline:** The full name of the operating airline.
- **FlightNum:** The flight number assigned by the airline.
- **TailNum:** The aircraft registration number (used to identify the specific aircraft).
- **ActualElapsedTime:** The total time (in minutes) between takeoff and landing.
- **CRSElapsedTime:** The scheduled flight duration in minutes.
- **AirTime:** The time spent in the air, excluding taxiing.
- **ArrDelay:** The difference (in minutes) between actual and scheduled arrival times. Positive values indicate delays.
- **DepDelay:** The difference (in minutes) between actual and scheduled departure times.
- **Origin:** The three-letter IATA code of the departure airport.

- **Org_Airport:** The full name of the origin airport.
- **Dest:** The three-letter IATA code of the destination airport.
- **Dest_Airport:** The full name of the destination airport.
- **Distance:** The distance between origin and destination airports (in miles).
- **TaxiIn:** Time spent taxiing from the runway to the gate after landing.
- **TaxiOut:** Time spent taxiing from the gate to the runway before takeoff.
- **Cancelled:** A binary indicator (1 = flight cancelled, 0 = not cancelled).
- **CancellationCode:** The reason for cancellation, such as weather, carrier, or security.
- **Diverted:** A binary indicator (1 = flight diverted to another airport, 0 = not diverted).
- **CarrierDelay:** Delay minutes caused by the airline's operations (e.g., crew or maintenance issues).
- **WeatherDelay:** Delay minutes caused by weather conditions.
- **NASDelay:** Delay caused by the National Airspace System (e.g., congestion or air traffic control).
- **SecurityDelay:** Delay caused by security checks or related incidents.
- **LateAircraftDelay:** Delay caused by a late incoming aircraft from a previous flight.

Dataset Purpose and Relevance

This dataset provides a rich foundation for analyzing and understanding **the key causes of flight delays** in the U.S. aviation industry. With over **480,000 flight records**, it offers sufficient data to explore:

- Delay frequency by airline, airport, and day of the week.
- Correlation between delay causes and total delay duration.
- The most frequent and impactful delay categories.

Such analysis helps reveal operational inefficiencies, seasonal trends, and patterns that directly affect flight punctuality and passenger satisfaction.

The dataset was carefully reviewed for consistency before analysis — ensuring correct data types, valid time values, and meaningful numerical ranges. Missing or inconsistent records were handled during the **data cleaning phase**, which is detailed in the following section.

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Flight Delay and Causes

This Dataset contains Flights trip and multiple cause of delay

Data Card Code (5) Discussion (0) Suggestions (0)

About Dataset

Context

Using this data you can find what caused the delay for flight whether it's Security delay, NAS delay or Carrier delay, etc.

Content

What's inside is more than just rows and columns. Make it easy for others to get started by describing how you acquired the data and what time period it represents, too.

Data Dictionary

1. DayOfWeek → 1 (Monday) - 7 (Sunday)
2. Date → Scheduled date

Usability

8.53

License

Community Data License Agree...

Expected update frequency

Not specified

Tags

Tabular Beginner

Exploratory Data Analysis

DayOfWeel	Date	DepTim	ArrTim	CRSArrTim	UniqueCarrie	Airline	FlightNun	TailNun	ActualElapsedTim	CRSElapsedTim	AirTim	ArrDela	DepDela	Origin	Org_Airpor	Dest	Dest_Airpor	Distanc	TaxiIn
4	3/1/2019	1829	1959	1925	WN	Southwes	3920	N464WN	90	90	77	34	34	IND	Indianapolis In	BWI	Baltimore-Wasl	515	3
4	3/1/2019	1937	2037	1940	WN	Southwes	509	N763SW	240	250	230	57	67	IND	Indianapolis In	LAS	McCarran Inter	1591	3
4	3/1/2019	1644	1845	1725	WN	Southwes	1333	N334SW	121	135	107	80	94	IND	Indianapolis In	MCO	Orlando Interna	828	6
4	3/1/2019	1452	1640	1625	WN	Southwes	675	N286WN	228	240	213	15	27	IND	Indianapolis In	PHX	Phoenix Sky Ha	1489	7
4	3/1/2019	1323	1526	1510	WN	Southwes	4	N674AA	123	135	110	16	28	IND	Indianapolis In	TPA	Tampa Internat	838	4
4	3/1/2019	1416	1512	1435	WN	Southwes	54	N643SW	56	70	49	37	51	ISP	Long Island Ma	BWI	Baltimore-Wasl	220	2
4	3/1/2019	1657	1754	1735	WN	Southwes	623	N724SW	57	70	47	19	32	ISP	Long Island Ma	BWI	Baltimore-Wasl	220	5
4	3/1/2019	1422	1657	1610	WN	Southwes	188	N215WN	155	195	143	47	87	ISP	Long Island Ma	FLL	Fort Lauderdale	1093	6
4	3/1/2019	2107	2334	2230	WN	Southwes	362	N798SW	147	165	134	64	82	ISP	Long Island Ma	MCO	Orlando Interna	972	6
4	3/1/2019	1812	1927	1815	WN	Southwes	422	N779SW	135	145	118	72	82	ISP	Long Island Ma	MDW	Chicago Midwa	765	6
4	3/1/2019	1326	1559	1530	WN	Southwes	1056	N459WN	153	180	143	29	56	ISP	Long Island Ma	PBI	Palm Beach Int	1052	5
4	3/1/2019	1450	1806	1745	WN	Southwes	3244	N475WN	136	130	121	21	15	JAN	Jackson-Evers	BWI	Baltimore-Wasl	888	7
4	3/1/2019	2245	2354	1850	WN	Southwes	186	N792SW	69	80	59	304	315	JAN	Jackson-Evers	HOU	William P. Hobl	359	3
4	3/1/2019	2025	2135	2100	WN	Southwes	3154	N252WN	70	80	60	35	45	JAN	Jackson-Evers	HOU	William P. Hobl	359	3
4	3/1/2019	1038	1314	1225	WN	Southwes	1035	N346SW	96	100	81	49	53	JAN	Jackson-Evers	MCO	Orlando Interna	587	8
4	3/1/2019	1900	2123	2045	WN	Southwes	205	N299WN	143	115	97	38	10	JAN	Jackson-Evers	MDW	Chicago Midwa	666	40
4	3/1/2019	948	959	940	WN	Southwes	3430	N487WN	71	75	59	19	23	JAX	Jacksonville Int	BHM	Birmingham-Sh	365	3
4	3/1/2019	646	725	655	WN	Southwes	1580	N243WN	99	95	77	30	26	JAX	Jacksonville Int	BNA	Nashville Interr	484	6
4	3/1/2019	1110	1136	1110	WN	Southwes	2195	N479WN	86	90	72	26	30	JAX	Jacksonville Int	BNA	Nashville Interr	484	5
4	3/1/2019	1813	1936	1905	WN	Southwes	54	N643SW	143	150	125	31	38	JAX	Jacksonville Int	HOU	William P. Hobl	816	6
4	3/1/2019	1734	1941	1905	WN	Southwes	23	N521SW	127	135	113	36	44	JAX	Jacksonville Int	PHL	Philadelphia Int	742	4
4	3/1/2019	958	1052	950	WN	Southwes	1574	N791SW	54	50	36	62	58	JAX	Jacksonville Int	TPA	Tampa Internat	180	4
4	3/1/2019	1538	1753	1710	WN	Southwes	500	N799SW	75	85	60	43	53	LAS	McCarran Inte	ABQ	Albuquerque In	487	4
4	3/1/2019	2248	102	2345	WN	Southwes	890	N618WN	74	80	60	77	83	LAS	McCarran Inte	ABQ	Albuquerque In	487	4
4	3/1/2019	1327	1550	1500	WN	Southwes	1171	N682SW	83	90	65	50	57	LAS	McCarran Inte	ABQ	Albuquerque In	487	3
4	3/1/2019	1832	148	30	WN	Southwes	302	N473WN	256	275	243	78	97	LAS	McCarran Inte	ALB	Albany Internat	2237	3
4	3/1/2019	1229	1633	1555	WN	Southwes	1079	N351SW	124	120	91	38	34	LAS	McCarran Inte	AMA	Rick Husband A	758	5
4	3/1/2019	2118	144	50	WN	Southwes	2021	N499WN	146	155	127	54	63	LAS	McCarran Inte	AUS	Austin-Bergstrc	1090	5
4	3/1/2019	1739	114	25	WN	Southwes	1018	N245WN	275	285	253	49	59	LAS	McCarran Inte	BDL	Bradley Interna	2298	9
4	3/1/2019	1325	1841	1810	WN	Southwes	419	N275WN	196	210	178	31	45	LAS	McCarran Inte	BNA	Nashville Interr	1588	6
4	3/1/2019	1506	2030	2010	WN	Southwes	2032	N271WN	204	210	183	20	26	LAS	McCarran Inte	BNA	Nashville Interr	1588	6
4	3/1/2019	2039	155	55	WN	Southwes	3940	N434WN	196	205	177	60	69	LAS	McCarran Inte	BNA	Nashville Interr	1588	5
4	3/1/2019	1611	1849	1825	WN	Southwes	538	N619SW	98	110	73	24	36	LAS	McCarran Inte	BOI	Boise AirportÂ	520	3
4	3/1/2019	1824	117	25	WN	Southwes	2383	N290WN	233	250	221	52	69	LAS	McCarran Inte	BUF	Buffalo Niagara	1987	2
4	3/1/2019	2118	2224	2115	WN	Southwes	219	N383SW	66	60	46	69	63	LAS	McCarran Inte	BUR	Bob Hope Airpo	223	3

T	U	V	W	X	Y	Z	AA	AB	AC
TaxiIn	TaxiOu	Cancellec	CancellationCod	Diverte	CarrierDela	WeatherDela	NASDelay	SecurityDela	LateAircraftDela
3	10	0	N	0	2	0	0	0	32
3	7	0	N	0	10	0	0	0	47
6	8	0	N	0	8	0	0	0	72
7	8	0	N	0	3	0	0	0	12
4	9	0	N	0	0	0	0	0	16
2	5	0	N	0	12	0	0	0	25
5	5	0	N	0	7	0	0	0	12
6	6	0	N	0	40	0	0	0	7
6	7	0	N	0	5	0	0	0	59
6	11	0	N	0	3	0	0	0	69
5	5	0	N	0	0	0	0	0	29
7	8	0	N	0	0	0	6	0	15
3	7	0	N	0	282	0	0	0	22
3	7	0	N	0	26	0	0	0	9
8	7	0	N	0	7	0	0	0	42
40	6	0	N	0	1	0	28	0	9
3	9	0	N	0	0	0	0	0	19
6	16	0	N	0	26	0	4	0	0
5	9	0	N	0	0	0	0	10	16
6	12	0	N	0	11	0	0	0	20
4	10	0	N	0	3	0	0	0	33
4	14	0	N	0	0	0	4	0	58
4	11	0	N	0	15	0	0	0	28
4	10	0	N	0	7	0	0	0	70
3	15	0	N	0	50	0	0	0	0
3	10	0	N	0	8	0	0	0	70
5	28	0	N	0	3	0	4	0	31
5	14	0	N	0	23	0	0	0	31
9	13	0	N	0	19	0	0	0	30
6	12	0	N	0	2	0	0	0	29
6	15	0	N	0	18	0	0	0	2
5	14	0	N	0	0	0	22	0	38
3	22	0	N	0	9	0	0	0	15
2	10	0	N	0	48	0	0	0	4
3	17	0	N	0	17	0	6	0	46

3.2 Data Cleaning

Before performing any analysis, the dataset underwent a comprehensive data cleaning process to ensure accuracy, consistency, and reliability of the results. Raw aviation data often contains inconsistencies, missing values, and format errors due to the large volume of records collected from multiple sources. Therefore, the cleaning phase was a critical step in preparing the dataset for analysis.

Handling Missing Values

One of the main issues identified during the data inspection phase was the presence of a large number of **null (missing)** values in two specific columns:

- **Org_Airport** (Origin Airport Name)
- **Dest_Airport** (Destination Airport Name)

These two columns represent the full names of the departure and destination airports, respectively. The missing values in these fields could affect the accuracy of analyses related to route performance and airport-specific delay trends.

To resolve this issue, the approach used was both **systematic and data-driven**:

1. Each flight record also contained the **Origin** and **Dest** columns, which store the **IATA airport codes** (three-letter abbreviations such as *JFK*, *LAX*, *ORD*, etc.).
2. These IATA codes were used as reliable identifiers for the airports.
3. Using these codes, an external lookup process was performed to retrieve the corresponding full airport names.
4. This lookup was done through **ChatGPT**, which was used to query the **official airport directory** and other reliable aviation data sources to match each missing airport code with its correct name.
5. Once the full airport names were identified, they were inserted back into the dataset to replace the null values in both *Org_Airport* and *Dest_Airport* columns.

This method ensured 100% consistency between the airport codes and their names, and eliminated any ambiguity in route-level analysis. After the replacement process, the dataset was rechecked to confirm that no null values remained in these columns.

Fixing Date Format Issues

Another major issue in the raw dataset was related to the **date formatting inconsistency**.
The column **Date** (flight date) was found to have its **month and day components reversed**, which caused incorrect interpretation of flight dates. For instance, a flight recorded as 05/12/2019 was interpreted as **May 12th, 2019**, while the correct date was **December 5th, 2019**.

To correct this issue, the following steps were performed using **Microsoft Excel**:

1. The **Text to Columns** feature was used to separate the date elements (day, month, year).
2. The components were then rearranged in the correct order — ensuring the format followed **DD/MM/YYYY** instead of **MM/DD/YYYY**.
3. After reformatting, the separated columns were merged again into a single unified date column using Excel's date conversion functions.
4. The corrected date column was verified across multiple records to ensure consistent interpretation across all entries.

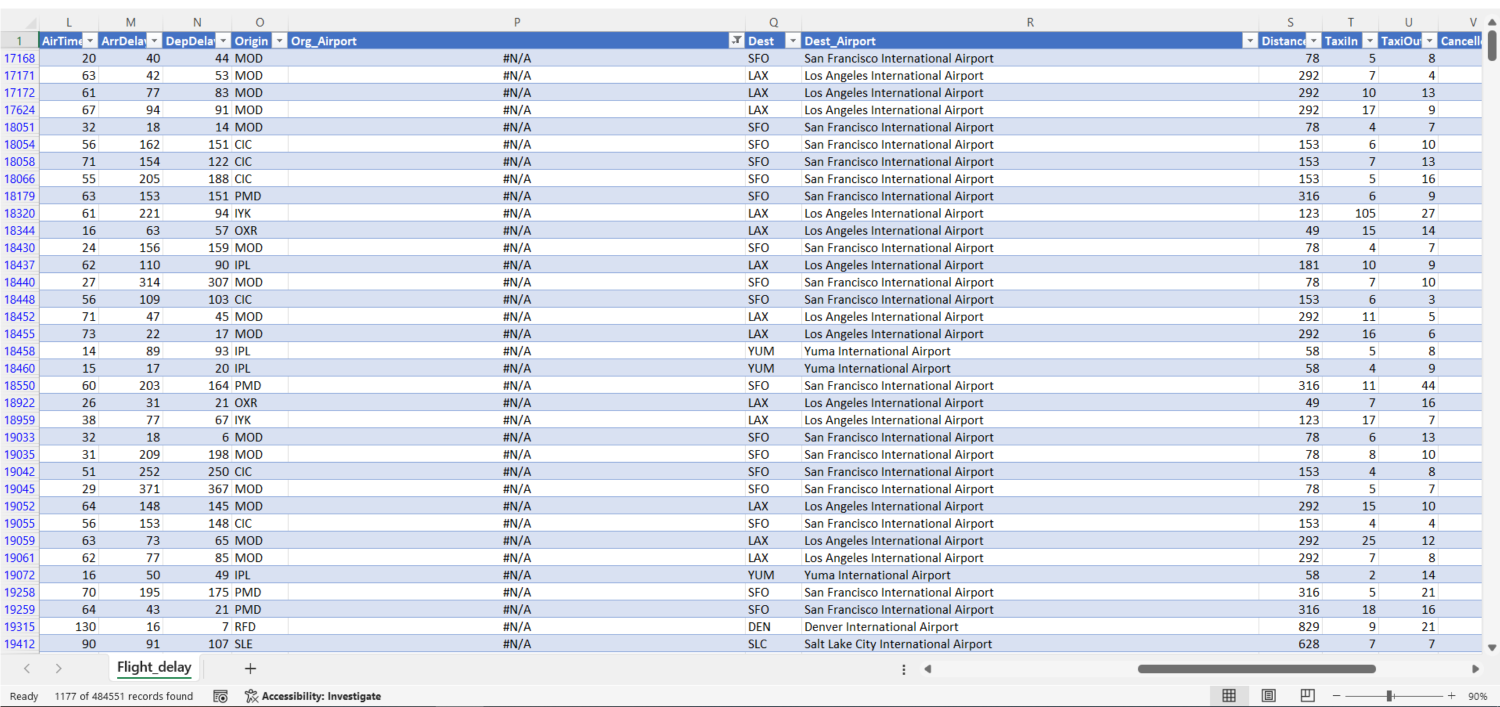
This correction was crucial because accurate date values are essential for performing **time-series analysis** and identifying **delay patterns by day, month, or weekday**.

Final Data Quality Check

After completing the above cleaning steps, the dataset was revalidated to confirm:

- No remaining null or invalid entries in the airport columns.
- All dates were properly standardized and correctly ordered.
- Column data types (numeric, text, date) were consistent and ready for further processing.

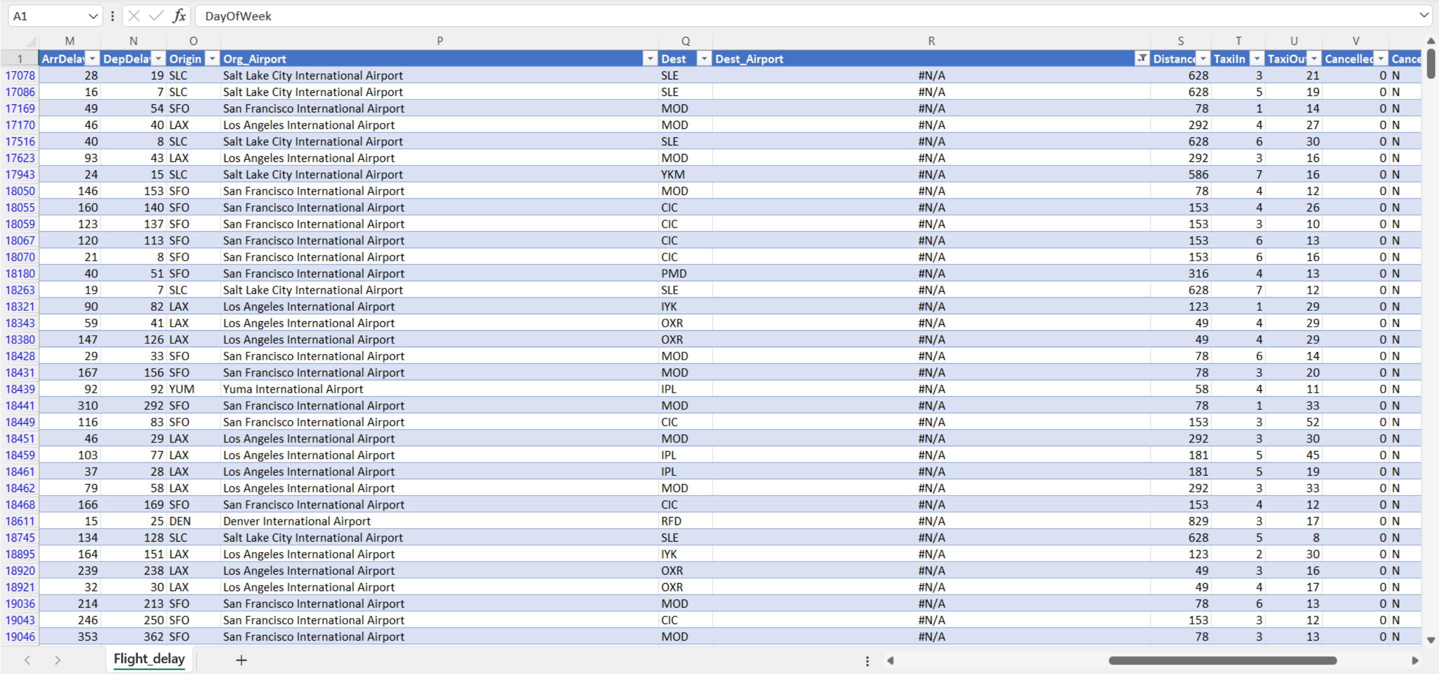
As a result of this cleaning phase, the dataset became fully structured, reliable, and ready for subsequent preprocessing and analysis tasks.



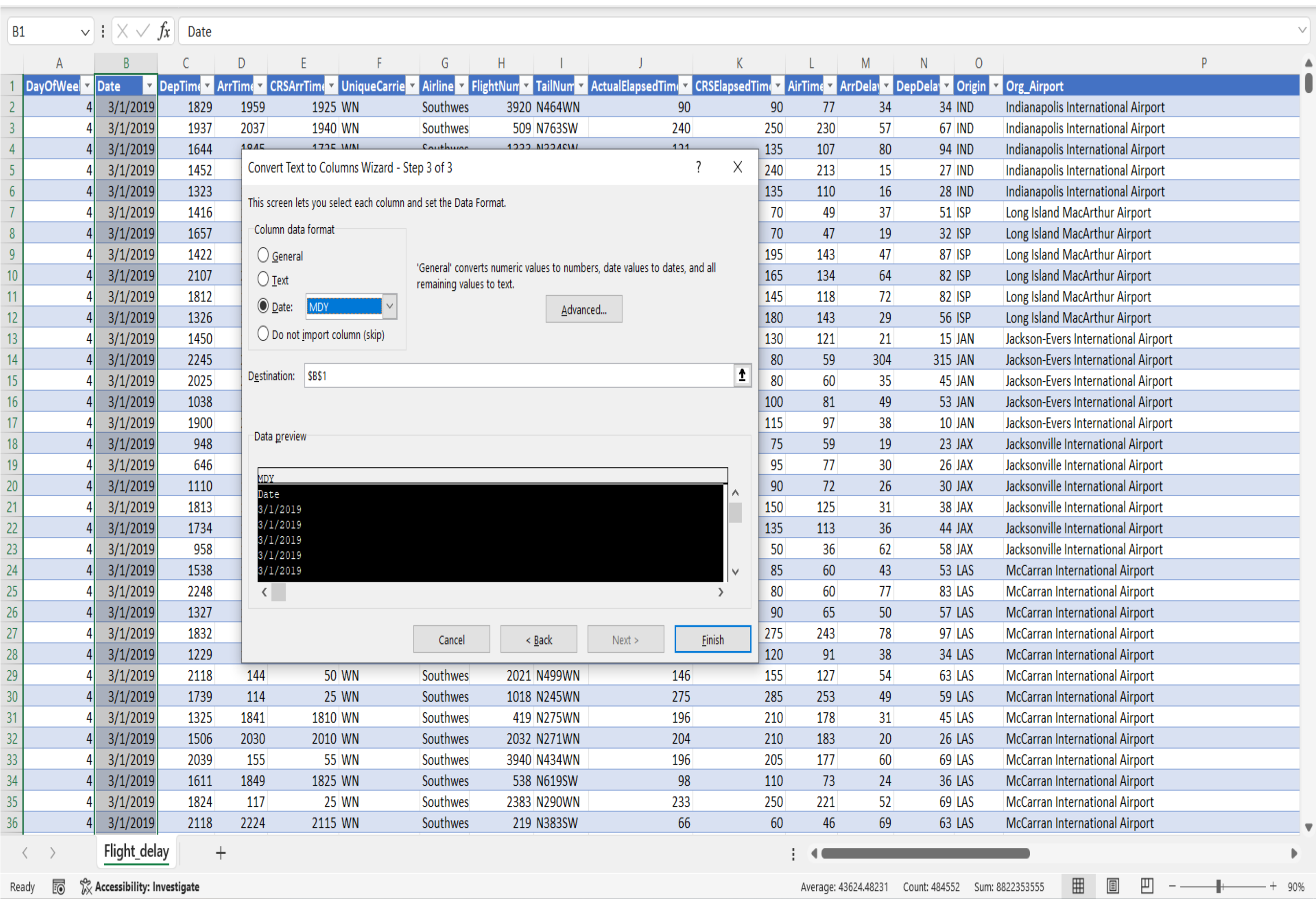
	L	M	N	O	P	Q	R	S	T	U	V
1	AirTime	ArrDela	DepDela	Origin	Org_Airport	Dest	Dest_Airport	Distanc	TaxiIn	TaxiOu	Cancell
17168	20	40	44	MOD	#N/A	SFO	San Francisco International Airport	78	5	8	
17171	63	42	53	MOD	#N/A	LAX	Los Angeles International Airport	292	7	4	
17172	61	77	83	MOD	#N/A	LAX	Los Angeles International Airport	292	10	13	
17624	67	94	91	MOD	#N/A	LAX	Los Angeles International Airport	292	17	9	
18051	32	18	14	MOD	#N/A	SFO	San Francisco International Airport	78	4	7	
18054	56	162	151	CIC	#N/A	SFO	San Francisco International Airport	153	6	10	
18058	71	154	122	CIC	#N/A	SFO	San Francisco International Airport	153	7	13	
18066	55	205	188	CIC	#N/A	SFO	San Francisco International Airport	153	5	16	
18179	63	153	151	PMD	#N/A	SFO	San Francisco International Airport	316	6	9	
18320	61	221	94	IYK	#N/A	LAX	Los Angeles International Airport	123	105	27	
18344	16	63	57	OXR	#N/A	LAX	Los Angeles International Airport	49	15	14	
18430	24	156	159	MOD	#N/A	SFO	San Francisco International Airport	78	4	7	
18437	62	110	90	IPL	#N/A	LAX	Los Angeles International Airport	181	10	9	
18440	27	314	307	MOD	#N/A	SFO	San Francisco International Airport	78	7	10	
18448	56	109	103	CIC	#N/A	SFO	San Francisco International Airport	153	6	3	
18452	71	47	45	MOD	#N/A	LAX	Los Angeles International Airport	292	11	5	
18455	73	22	17	MOD	#N/A	LAX	Los Angeles International Airport	292	16	6	
18458	14	89	93	IPL	#N/A	YUM	Yuma International Airport	58	5	8	
18460	15	17	20	IPL	#N/A	YUM	Yuma International Airport	58	4	9	
18550	60	203	164	PMD	#N/A	SFO	San Francisco International Airport	316	11	44	
18922	26	31	21	OXR	#N/A	LAX	Los Angeles International Airport	49	7	16	
18959	38	77	67	IYK	#N/A	LAX	Los Angeles International Airport	123	17	7	
19033	32	18	6	MOD	#N/A	SFO	San Francisco International Airport	78	6	13	
19035	31	209	198	MOD	#N/A	SFO	San Francisco International Airport	78	8	10	
19042	51	252	250	CIC	#N/A	SFO	San Francisco International Airport	153	4	8	
19045	29	371	367	MOD	#N/A	SFO	San Francisco International Airport	78	5	7	
19052	64	148	145	MOD	#N/A	LAX	Los Angeles International Airport	292	15	10	
19055	56	153	148	CIC	#N/A	SFO	San Francisco International Airport	153	4	4	
19059	63	73	65	MOD	#N/A	LAX	Los Angeles International Airport	292	25	12	
19061	62	77	85	MOD	#N/A	LAX	Los Angeles International Airport	292	7	8	
19072	16	50	49	IPL	#N/A	YUM	Yuma International Airport	58	2	14	
19258	70	195	175	PMD	#N/A	SFO	San Francisco International Airport	316	5	21	
19259	64	43	21	PMD	#N/A	SFO	San Francisco International Airport	316	18	16	
19315	130	16	7	RFD	#N/A	DEN	Denver International Airport	829	9	21	
19412	90	91	107	SLE	#N/A	SLC	Salt Lake City International Airport	628	7	7	

Flight_delay

Ready 1177 of 484551 records found Accessibility: Investigate 90%



	ArrDela	DepDela	Origin	Org_Airport	Dest	Dest_Airport	Distanc	TaxiIn	TaxiOu	Cancelle	Cance
17078	28	19	SLC	Salt Lake City International Airport	SLE	#N/A	628	3	21	0	N
17086	16	7	SLC	Salt Lake City International Airport	SLE	#N/A	628	5	19	0	N
17169	49	54	SFO	San Francisco International Airport	MOD	#N/A	78	1	14	0	N
17170	46	40	LAX	Los Angeles International Airport	MOD	#N/A	292	4	27	0	N
17516	40	8	SLC	Salt Lake City International Airport	SLE	#N/A	628	6	30	0	N
17623	93	43	LAX	Los Angeles International Airport	MOD	#N/A	292	3	16	0	N
17943	24	15	SLC	Salt Lake City International Airport	YKM	#N/A	586	7	16	0	N
18050	146	153	SFO	San Francisco International Airport	MOD	#N/A	78	4	12	0	N
18055	160	140	SFO	San Francisco International Airport	CIC	#N/A	153	4	26	0	N
18059	123	137	SFO	San Francisco International Airport	CIC	#N/A	153	3	10	0	N
18067	120	113	SFO	San Francisco International Airport	CIC	#N/A	153	6	13	0	N
18070	21	8	SFO	San Francisco International Airport	CIC	#N/A	153	6	16	0	N
18180	40	51	SFO	San Francisco International Airport	PMD	#N/A	316	4	13	0	N
18263	19	7	SLC	Salt Lake City International Airport	SLE	#N/A	628	7	12	0	N
18321	90	82	LAX	Los Angeles International Airport	IYK	#N/A	123	1	29	0	N
18343	59	41	LAX	Los Angeles International Airport	OXR	#N/A	49	4	29	0	N
18380	147	126	LAX	Los Angeles International Airport	OXR	#N/A	49	4	29	0	N
18428	29	33	SFO	San Francisco International Airport	MOD	#N/A	78	6	14	0	N
18431	167	156	SFO	San Francisco International Airport	MOD	#N/A	78	3	20	0	N
18439	92	92	YUM	Yuma International Airport	IPL	#N/A	58	4	11	0	N
18441	310	292	SFO	San Francisco International Airport	MOD	#N/A	78	1	33	0	N
18449	116	83	SFO	San Francisco International Airport	CIC	#N/A	153	3	52	0	N
18451	46	29	LAX	Los Angeles International Airport	MOD	#N/A	292	3	30	0	N
18459	103	77	LAX	Los Angeles International Airport	IPL	#N/A	181	5	45	0	N
18461	37	28	LAX	Los Angeles International Airport	IPL	#N/A	181	5	19	0	N
18462	79	58	LAX	Los Angeles International Airport	MOD	#N/A	292	3	33	0	N
18468	166	169	SFO	San Francisco International Airport	CIC	#N/A	153	4	12	0	N
18611	15	25	DEN	Denver International Airport	RFD	#N/A	829	3	17	0	N
18745	134	128	SLC	Salt Lake City International Airport	SLE	#N/A	628	5	8	0	N
18895	164	151	LAX	Los Angeles International Airport	IYK	#N/A	123	2	30	0	N
18920	239	238	LAX	Los Angeles International Airport	OXR	#N/A	49	3	16	0	N
18921	32	30	LAX	Los Angeles International Airport	OXR	#N/A	49	4	17	0	N
19036	214	213	SFO	San Francisco International Airport	MOD	#N/A	78	6	13	0	N
19043	246	250	SFO	San Francisco International Airport	CIC	#N/A	153	3	12	0	N
19046	353	362	SFO	San Francisco International Airport	MOD	#N/A	78	3	13	0	N



	DayOfWeel	Date	DepTim	ArrTim	CRSArrTim	UniqueCarrie	Airline	FlightNum	TailNum	ActualElapsedTim	CRSElapsedTim	AirTim	ArrDela	DepDela	Origin	Org_Airport
2	4	3/1/2019	1829	1959	1925	WN	Southwes	3920	N464WN	90	90	77	34	34	IND	Indianapolis International Airport
3	4	3/1/2019	1937	2037	1940	WN	Southwes	509	N763SW	240	250	230	57	67	IND	Indianapolis International Airport
4	4	3/1/2019	1644								135	107	80	94	IND	Indianapolis International Airport
5	4	3/1/2019	1452								240	213	15	27	IND	Indianapolis International Airport
6	4	3/1/2019	1323								135	110	16	28	IND	Indianapolis International Airport
7	4	3/1/2019	1416								70	49	37	51	ISP	Long Island MacArthur Airport
8	4	3/1/2019	1657								70	47	19	32	ISP	Long Island MacArthur Airport
9	4	3/1/2019	1422								195	143	47	87	ISP	Long Island MacArthur Airport
10	4	3/1/2019	2107								165	134	64	82	ISP	Long Island MacArthur Airport
11	4	3/1/2019	1812								145	118	72	82	ISP	Long Island MacArthur Airport
12	4	3/1/2019	1326								180	143	29	56	ISP	Long Island MacArthur Airport
13	4	3/1/2019	1450								130	121	21	15	JAN	Jackson-Evers International Airport
14	4	3/1/2019	2245								80	59	304	315	JAN	Jackson-Evers International Airport
15	4	3/1/2019	2025								80	60	35	45	JAN	Jackson-Evers International Airport
16	4	3/1/2019	1038								100	81	49	53	JAN	Jackson-Evers International Airport
17	4	3/1/2019	1900								115	97	38	10	JAN	Jackson-Evers International Airport
18	4	3/1/2019	948								75	59	19	23	JAX	Jacksonville International Airport
19	4	3/1/2019	646								95	77	30	26	JAX	Jacksonville International Airport
20	4	3/1/2019	1110								90	72	26	30	JAX	Jacksonville International Airport
21	4	3/1/2019	1813								150	125	31	38	JAX	Jacksonville International Airport
22	4	3/1/2019	1734								135	113	36	44	JAX	Jacksonville International Airport
23	4	3/1/2019	958								50	36	62	58	JAX	Jacksonville International Airport
24	4	3/1/2019	1538								85	60	43	53	LAS	McCarran International Airport
25	4	3/1/2019	2248								80	60	77	83	LAS	McCarran International Airport
26	4	3/1/2019	1327								90	65	50	57	LAS	McCarran International Airport
27	4	3/1/2019	1832								275	243	78	97	LAS	McCarran International Airport
28	4	3/1/2019	1229								120	91	38	34	LAS	McCarran International Airport
29	4	3/1/2019	2118	144	50	WN	Southwes	2021	N499WN	146	155	127	54	63	LAS	McCarran International Airport
30	4	3/1/2019	1739	114	25	WN	Southwes	1018	N245WN	275	285	253	49	59	LAS	McCarran International Airport
31	4	3/1/2019	1325	1841	1810	WN	Southwes	419	N275WN	196	210	178	31	45	LAS	McCarran International Airport
32	4	3/1/2019	1506	2030	2010	WN	Southwes	2032	N271WN	204	210	183	20	26	LAS	McCarran International Airport
33	4	3/1/2019	2039	155	55	WN	Southwes	3940	N434WN	196	205	177	60	69	LAS	McCarran International Airport
34	4	3/1/2019	1611	1849	1825	WN	Southwes	538	N619SW	98	110	73	24	36	LAS	McCarran International Airport
35	4	3/1/2019	1824	117	25	WN	Southwes	2383	N290WN	233	250	221	52	69	LAS	McCarran International Airport
36	4	3/1/2019	2118	2224	2115	WN	Southwes	219	N383SW	66	60	46	69	63	LAS	McCarran International Airport

3.3 Data Preprocessing

Definition and Purpose

Data preprocessing represents a fundamental phase in any data analysis or machine learning project. It refers to the set of techniques and transformations applied to raw data in order to make it suitable for analytical processing.

In aviation-related datasets, raw flight data often contains inconsistencies, missing entries, and structural limitations that can hinder accurate analysis. Preprocessing ensures that data is properly structured, standardized, and enriched with new calculated fields that add analytical value.

The goal of preprocessing in this project was to **enhance the flight delay dataset** for deeper insights — by cleaning, formatting, enriching, and restructuring it. This was done mainly through **Microsoft Excel** and **Power Query**, along with external lookups via **ChatGPT** for airport information.

1. General Enhancements

After the initial cleaning, the dataset was extended with **multiple derived columns**, standardizations, and logical transformations to support advanced analytical tasks such as:

- Delay cause categorization
- Temporal (time-based) analysis
- Geographical segmentation
- Performance comparison between airports and airlines

All new columns were implemented using **Power Query M Language** and Excel functions.

2. Detailed Explanation of Each Added Column

2.1 Departure_Time

Purpose: To convert the integer-based *DepTime* (e.g., 1530) into a proper time format for accurate analysis of departure punctuality.

Power Query Code:

```
Departure_Time = Time.From(#time(
    Number.IntegerDivide([DepTime],100),
    Number.Mod([DepTime],100),
```

```
    0))
```

Explanation:

- The code divides the integer *DepTime* into hours and minutes.
- This conversion allowed precise comparison between scheduled and actual departure times.
- It also helped in identifying peak hours and evaluating delay frequency by time of day.

2.2 Arrival_Time

Purpose: Similar to the departure time conversion, this field transforms *ArrTime* from numeric to time format to allow accurate computation of flight duration and arrival punctuality.

Power Query Code:

```
Arrival_Time = Time.From(#time(
    Number.IntegerDivide([ArrTime],100),
    Number.Mod([ArrTime],100),
    0))
```

Analytical Value:
This transformation made it possible to calculate metrics like:

- Average arrival delay per airline.
- Correlation between departure and arrival delay patterns.
- Identification of late-night or early-morning delay trends.

2.3 CRS_Arrival_Time

Purpose: Convert the scheduled arrival time (*CRSArrTime*) into time format for direct comparison with actual arrival times.

Power Query Code:

```
CRS_Arrival_Time = Time.From(#time(
    Number.IntegerDivide([CRSArrTime],100),
    Number.Mod([CRSArrTime],100),
    0))
```

Outcome:
This column allowed the calculation of **planned vs actual** performance gaps, helping visualize scheduling accuracy and airline reliability.

2.4 Route

Purpose: To create a single descriptive column that defines each flight route as *"Origin – Destination"*.

Power Query Code:

```
Route = [Origin] & " - " & [Dest]
```

Analytical Benefit:

- Simplified route-based grouping and filtering.
- Enabled route-level delay comparisons (e.g., which routes are most delayed).
- Used directly in dashboards for route-level KPIs.

2.5 TotalDelay

Purpose: To calculate the total number of delay minutes per flight by summing all five delay categories.

Power Query Code:

```
TotalDelay = [CarrierDelay] + [WeatherDelay] +
             [NASDelay] + [SecurityDelay] + [LateAircraftDelay]
```

Analytical Value:
This aggregate measure allowed the calculation of:

- Total delay distribution across airlines.
- Ranking of most delayed routes or airports.
- Correlation between total delay and distance or time of day.

2.6 Main_Delay

Purpose: Identify **the main cause** of delay for each flight (i.e., which delay type contributed the most).

Power Query Code:

```
Main_Delay = let
    delays = [
        CarrierDelay = [CarrierDelay],
        WeatherDelay = [WeatherDelay],
        NASDelay = [NASDelay],
        SecurityDelay = [SecurityDelay],
        LateAircraftDelay = [LateAircraftDelay]
    ],
    maxValue = List.Max(Record.ToList(delays)),
    maxColumn =
List.First(List.Select(Record.FieldNames(delays),
                    (col) => Record.Field(delays, col) =
maxValue))
in
    maxColumn
```

Explanation:
This code dynamically checks which of the five delay columns has the highest value for each record and assigns its name as the dominant cause.
Analytical Impact:

- Simplified cause-based grouping.
- Helped determine that **LateAircraftDelay** was the most frequent main cause across the dataset.

2.7 Delay Ratios

Purpose: To calculate the percentage contribution of each delay type to the total delay per flight.

Power Query Code:

```
let
    TotalDelay = [CarrierDelay] + [WeatherDelay] + [NASDelay] +
[SecurityDelay] + [LateAircraftDelay],
    CarrierRatio = if TotalDelay > 0 then
Text.From(Number.Round([CarrierDelay] / TotalDelay * 100, 1)) &
"%" else "0%",
```

```
    WeatherRatio = if TotalDelay > 0 then
Text.From(Number.Round([WeatherDelay] / TotalDelay * 100, 1)) &
"%" else "0%",
    NASRatio = if TotalDelay > 0 then
Text.From(Number.Round([NASDelay] / TotalDelay * 100, 1)) & "%"
else "0%",
    SecurityRatio = if TotalDelay > 0 then
Text.From(Number.Round([SecurityDelay] / TotalDelay * 100, 1)) &
"%" else "0%",
    LateAircraftRatio = if TotalDelay > 0 then
Text.From(Number.Round([LateAircraftDelay] / TotalDelay * 100,
1)) & "%" else "0%"
in
    [CarrierRatio = CarrierRatio, WeatherRatio = WeatherRatio,
     NASRatio = NASRatio, SecurityRatio = SecurityRatio,
LateAircraftRatio = LateAircraftRatio]
```

Analytical Importance:

- Provided a proportional breakdown of delays per flight.
- Helped visualize which delay types dominate specific airlines or airports.
- Useful for pie chart and stacked bar representations in dashboards.

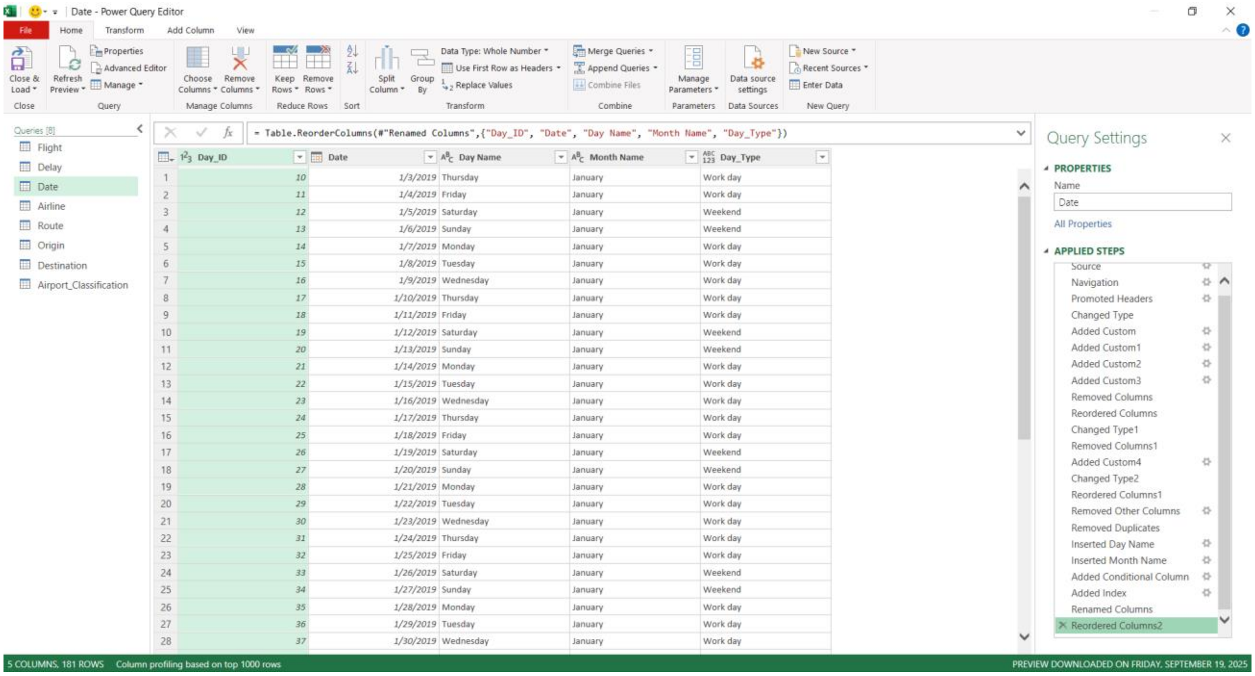
2.8 DayName & MonthName

Purpose: To enhance time-series analysis by labeling each record with its day of the week and month name.

Implementation: Created in Excel via *Insert Day Name* and *Insert Month Name* functions based on the Date column.

Analytical Role:

- Enabled aggregation of delays by weekday and month.
- Revealed seasonal and weekly patterns (e.g., Friday having higher delays).



2.9 City & State Columns

Purpose: To enrich geographical context for both origin and destination airports.

Columns Added:

- Origin_City, Origin_State, Dest_City, Dest_State

Method:
These fields were generated using **airport lookup data** obtained from verified online directories (via ChatGPT referencing official IATA sources).

Benefit:

- Enabled analysis of delays by region or state.
- Allowed creation of heat maps showing average delays across U.S. states.

2.10 Airport Category & Description

Purpose: To classify each airport according to its size and operational importance (Large, Medium, or Small Hub).

Implementation:
Added manually based on the Federal Aviation Administration (FAA) airport classification system.

Description Field:
A supplementary text column explaining the meaning of each category, for instance:

- *Large Hub:* handles ≥1% of total U.S. passenger enplanements.
- *Medium Hub:* handles between 0.25%–1%.
- *Small Hub:* less than 0.25%.

Analytical Purpose:

- Helped study whether larger airports face more congestion and delays.
- Supported segmentation in dashboards comparing delay averages by airport size.

2.11 Index Column

Purpose:
After removing duplicate rows, a new **Index** column was created to ensure each record is uniquely identified.

Implementation:
Automatically generated using Power Query's *Add Index Column* function.

Analytical Use:

- Essential for database joins and modeling in Power BI.
- Ensured no data duplication when merging tables in later modeling stages.

2.12 Status (Conditional Classification Column)

Purpose:
The **Status** column was created to categorize each flight based on its **arrival delay time** into one of three meaningful groups — *Early*, *On Time*, or *Delayed*.
This classification helps in simplifying interpretation of delay data and supports dashboard visualizations showing punctuality rates for each airline or airport.

Implementation:
The column was created in **Power Query** using the *Add Conditional Column* feature with the following logic:

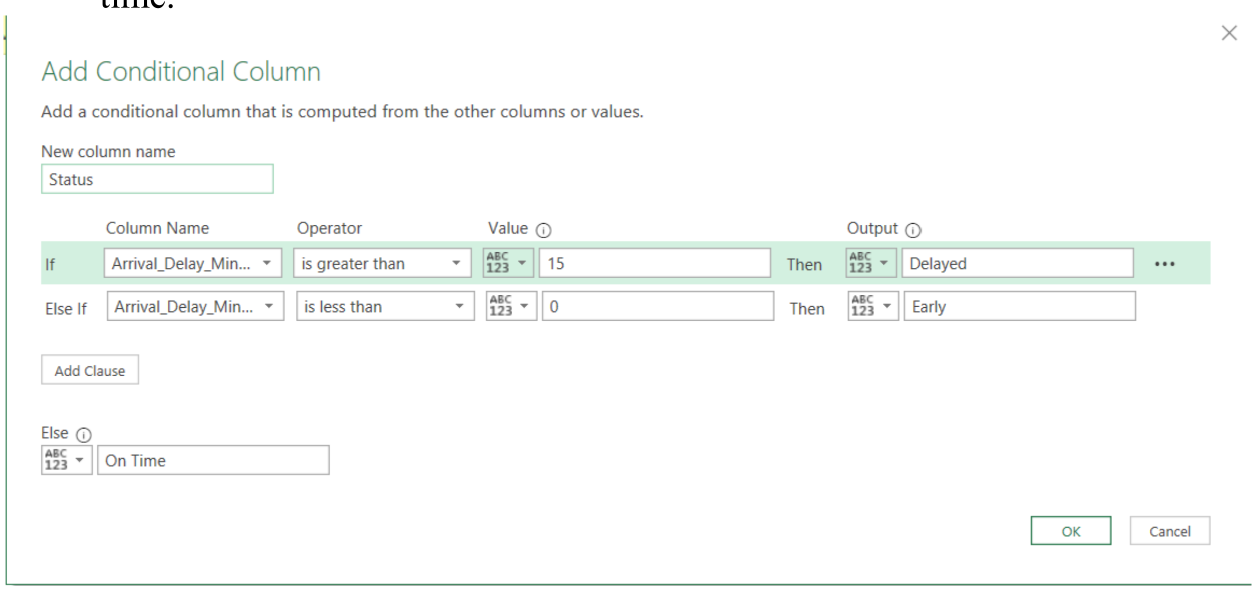
Condition	Description	Output
If [Arrival_Delay_Min] > 15	The flight arrived more than 15 minutes late	**"Delayed"**
Else if [Arrival_Delay_Min] < 0	The flight arrived earlier than scheduled	**"Early"**
Else	The flight arrived on schedule (within 15 minutes window)	**"On Time"**

Power Query M Code (Generated Automatically):

```
Status =
if [Arrival_Delay_Min] > 15 then "Delayed"
else if [Arrival_Delay_Min] < 0 then "Early"
else "On Time"
```

Explanation and Analytical Value:

- This transformation converts raw numerical delay data into categorical performance indicators.
- Airlines and airports can easily track what percentage of their flights are delayed, on time, or early.
- It's particularly useful in visual dashboards — for example, creating pie charts showing punctuality rates by airline or airport.
- The 15-minute threshold aligns with **U.S. Department of Transportation (DOT)** standards, which classify flights arriving within 15 minutes of scheduled time as "on time."



3. Columns Removed

To maintain dataset efficiency and clarity, the following were excluded:

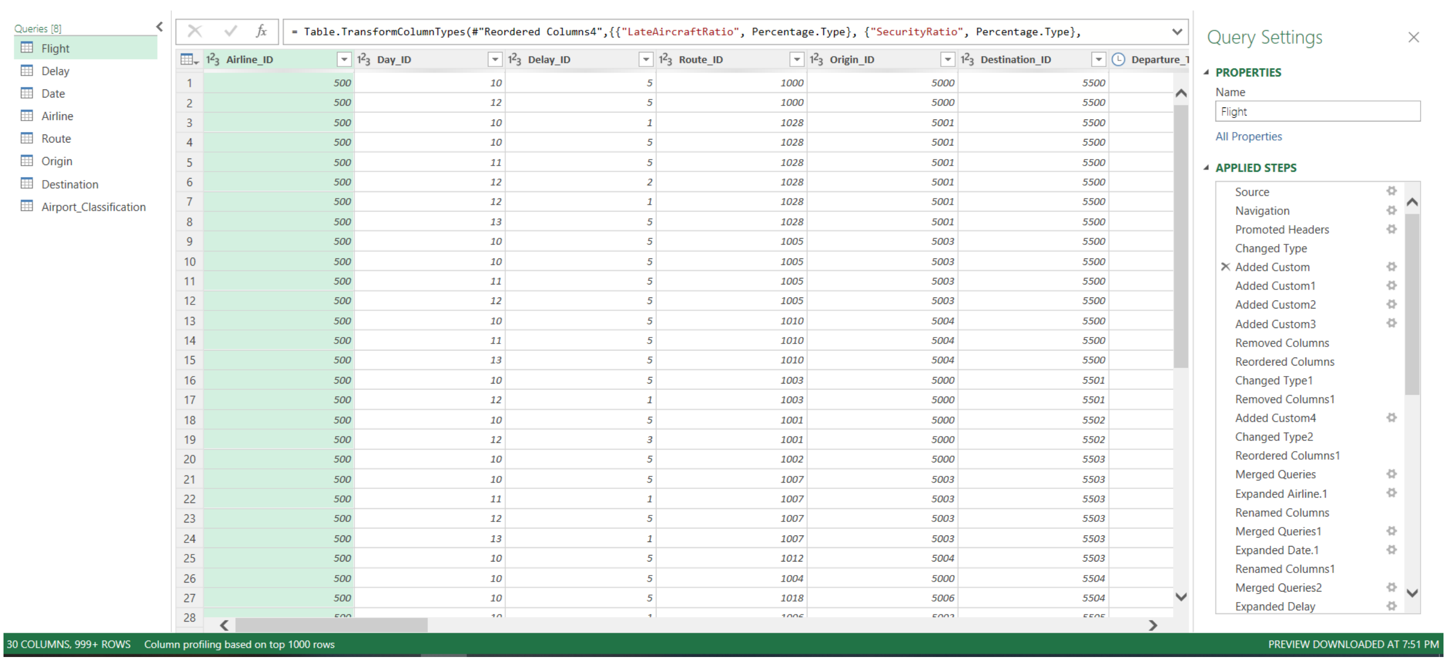
- **TailNum, FlightNum:** identifiers without analytical relevance.

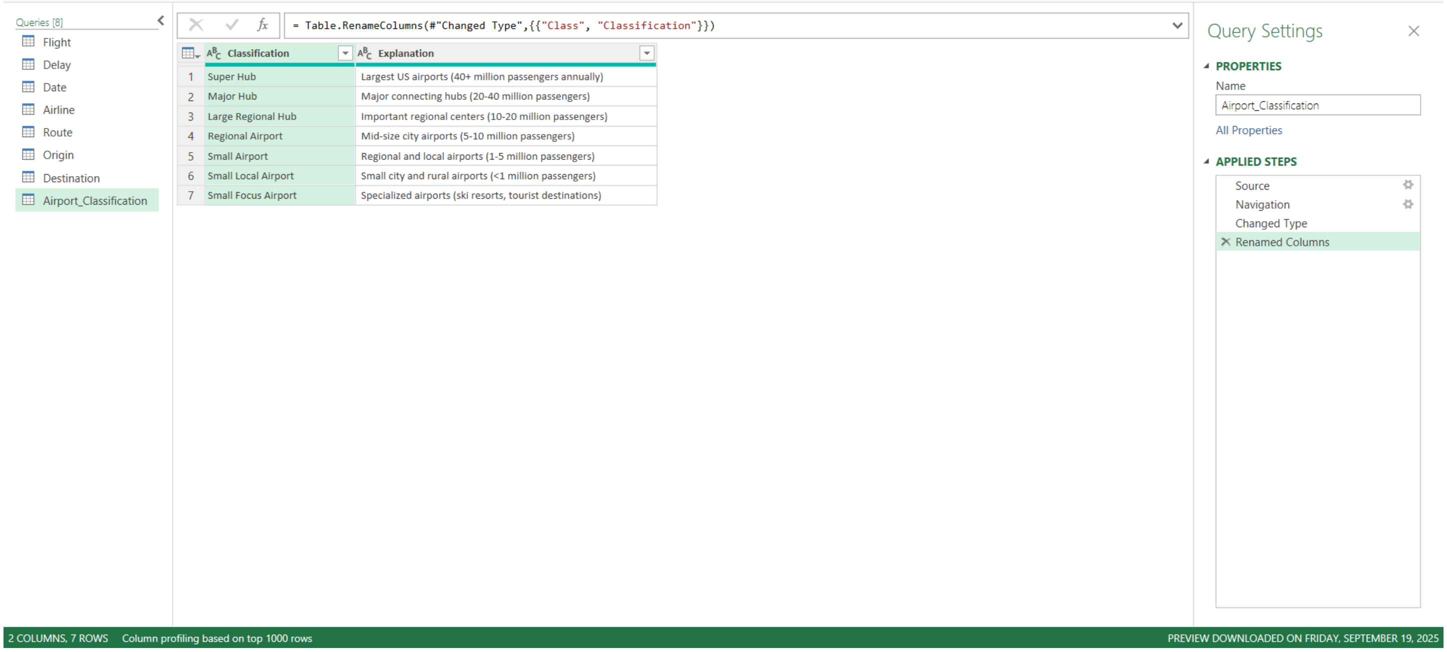
- **Cancelled, CancellationCode, Diverted:** filtered out to focus on operational flights only.
- Any columns containing static or repeated values that did not add value to analysis.

4. Analytical Impact of Preprocessing

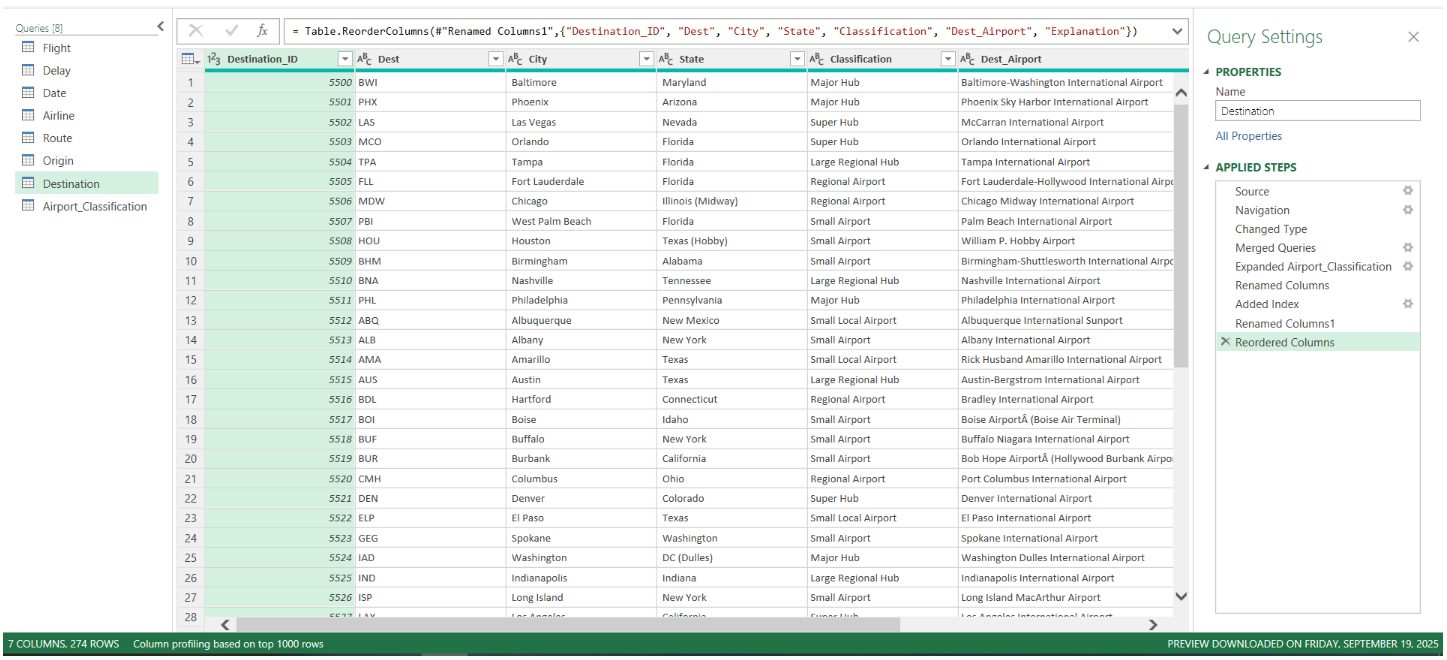
Through all these transformations, the dataset evolved from a raw structured file into a **powerful analytical model**.
Each new feature provided a new analytical dimension — time, geography, or cause — which allowed for:

- Delay pattern visualization by month, day, airline, and airport size.
- Quantitative understanding of major delay causes.
- Integration-ready dataset for advanced modeling and dashboard design.

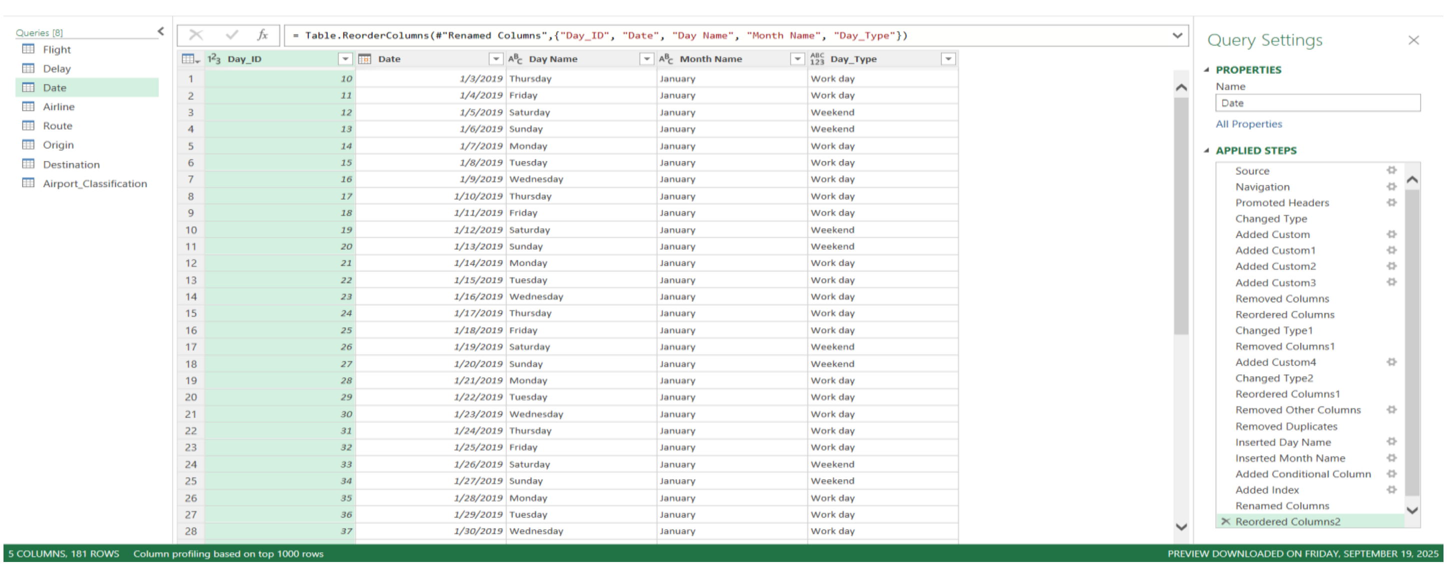




	Classification	Explanation
1	Super Hub	Largest US airports (40+ million passengers annually)
2	Major Hub	Major connecting hubs (20-40 million passengers)
3	Large Regional Hub	Important regional centers (10-20 million passengers)
4	Regional Airport	Mid-size city airports (5-10 million passengers)
5	Small Airport	Regional and local airports (1-5 million passengers)
6	Small Local Airport	Small city and rural airports (<1 million passengers)
7	Small Focus Airport	Specialized airports (ski resorts, tourist destinations)



	Destination_ID	Dest	City	State	Classification	Dest_Airport
1	5500	BWI	Baltimore	Maryland	Major Hub	Baltimore-Washington International Airport
2	5501	PHX	Phoenix	Arizona	Major Hub	Phoenix Sky Harbor International Airport
3	5502	LAS	Las Vegas	Nevada	Super Hub	McCarran International Airport
4	5503	MCO	Orlando	Florida	Super Hub	Orlando International Airport
5	5504	TPA	Tampa	Florida	Large Regional Hub	Tampa International Airport
6	5505	FLL	Fort Lauderdale	Florida	Regional Airport	Fort Lauderdale-Hollywood International Airpc
7	5506	MDW	Chicago	Illinois (Midway)	Regional Airport	Chicago Midway International Airport
8	5507	PBI	West Palm Beach	Florida	Small Airport	Palm Beach International Airport
9	5508	HOU	Houston	Texas (Hobby)	Small Airport	William P. Hobby Airport
10	5509	BHM	Birmingham	Alabama	Small Airport	Birmingham-Shuttlesworth International Airpc
11	5510	BNA	Nashville	Tennessee	Large Regional Hub	Nashville International Airport
12	5511	PHL	Philadelphia	Pennsylvania	Major Hub	Philadelphia International Airport
13	5512	ABQ	Albuquerque	New Mexico	Small Local Airport	Albuquerque International Sunport
14	5513	ALB	Albany	New York	Small Airport	Albany International Airport
15	5514	AMA	Amarillo	Texas	Small Local Airport	Rick Husband Amarillo International Airport
16	5515	AUS	Austin	Texas	Large Regional Hub	Austin-Bergstrom International Airport
17	5516	BDL	Hartford	Connecticut	Regional Airport	Bradley International Airport
18	5517	BOI	Boise	Idaho	Small Airport	Boise AirportÂ (Boise Air Terminal)
19	5518	BUF	Buffalo	New York	Small Airport	Buffalo Niagara International Airport
20	5519	BUR	Burbank	California	Small Airport	Bob Hope AirportÂ (Hollywood Burbank Airpo
21	5520	CMH	Columbus	Ohio	Regional Airport	Port Columbus International Airport
22	5521	DEN	Denver	Colorado	Super Hub	Denver International Airport
23	5522	ELP	El Paso	Texas	Small Local Airport	El Paso International Airport
24	5523	GEG	Spokane	Washington	Small Airport	Spokane International Airport
25	5524	IAD	Washington	DC (Dulles)	Major Hub	Washington Dulles International Airport
26	5525	IND	Indianapolis	Indiana	Large Regional Hub	Indianapolis International Airport
27	5526	ISP	Long Island	New York	Small Airport	Long Island MacArthur Airport
28	5527	LAX	Los Angeles	California	Super Hub	Los Angeles International Airport



	Day_ID	Date	Day Name	Month Name	Day_Type
1	10	1/3/2019	Thursday	January	Work day
2	11	1/4/2019	Friday	January	Work day
3	12	1/5/2019	Saturday	January	Weekend
4	13	1/6/2019	Sunday	January	Weekend
5	14	1/7/2019	Monday	January	Work day
6	15	1/8/2019	Tuesday	January	Work day
7	16	1/9/2019	Wednesday	January	Work day
8	17	1/10/2019	Thursday	January	Work day
9	18	1/11/2019	Friday	January	Work day
10	19	1/12/2019	Saturday	January	Weekend
11	20	1/13/2019	Sunday	January	Weekend
12	21	1/14/2019	Monday	January	Work day
13	22	1/15/2019	Tuesday	January	Work day
14	23	1/16/2019	Wednesday	January	Work day
15	24	1/17/2019	Thursday	January	Work day
16	25	1/18/2019	Friday	January	Work day
17	26	1/19/2019	Saturday	January	Weekend
18	27	1/20/2019	Sunday	January	Weekend
19	28	1/21/2019	Monday	January	Work day
20	29	1/22/2019	Tuesday	January	Work day
21	30	1/23/2019	Wednesday	January	Work day
22	31	1/24/2019	Thursday	January	Work day
23	32	1/25/2019	Friday	January	Work day
24	33	1/26/2019	Saturday	January	Weekend
25	34	1/27/2019	Sunday	January	Weekend
26	35	1/28/2019	Monday	January	Work day
27	36	1/29/2019	Tuesday	January	Work day
28	37	1/30/2019	Wednesday	January	Work day

3.4 Data Modeling

Data modeling is a critical phase in the analytical pipeline that transforms large and unstructured datasets into a well-organized data warehouse model. This step enables efficient querying, relationship building, and insightful visualization. The modeling process helps in structuring the dataset in a way that reduces redundancy, improves performance, and ensures consistency across analytical dimensions.

In this project, the original dataset consisted of a single large table containing over 480,000 flight records. Although this flat table contained all necessary information, its structure was not optimized for analysis. Therefore, it was crucial to **design a star schema model** that separates the data into a central **fact table** and multiple **dimension tables**.

4.1 Objective of Modeling

The primary goal of the modeling phase was to:

- Organize the cleaned dataset into a relational structure that supports fast querying and efficient dashboard creation.
- Eliminate data duplication and inconsistencies found in the raw dataset.
- Enable easy cross-analysis between different dimensions such as airline, route, airport, and time.
- Enhance analytical flexibility, allowing calculation of key metrics such as average delay per airline, delay cause breakdown, and on-time performance by airport type.

4.2 Transition from Preprocessing to Modeling

The preprocessing stage prepared the dataset by:

- Correcting data formats and null values.
- Adding analytical columns such as *TotalDelay*, *Main_Delay*, *Status*, *DayName*, and *MonthName*.
- Creating unique index columns for every logical group of information (airline, origin, destination, route, date, delay cause).

These unique **index keys** were the foundation for building relationships between tables in the data model.
Without this preparation, it would have been impossible to build a stable model or connect the tables correctly.

4.3 Rationale for Data Warehousing and Star Schema Design

Given the dataset's large size and the repetition of information across flights (e.g., the same airline, route, and airports repeating thousands of times), it was natural and efficient to move toward a **data warehouse design**.

A **star schema** was chosen because:

- It provides a simple, intuitive structure for analytical tools like Power BI or Excel Power Pivot.
- It clearly separates **facts (quantitative data)** from **dimensions (descriptive data)**.
- It supports **one-to-many relationships**, ensuring faster aggregation and simplified data relationships.

The **Flight table** serves as the **Fact table**, while six **Dimension tables** were derived from it. Each dimension represents a unique aspect of the flight operation (Airline, Route, Origin, Destination, Date, Delay).

4.4 Modeling Process and Challenges

Initially, when the dataset was divided into separate dimension tables, we applied **Remove Duplicates** to create unique values per dimension. However, this caused significant data loss during early merges because:

- Some keys were not unique.
- Merges between tables caused row mismatches.
- Duplicated routes or airports generated inflated record counts.

When we first merged the tables again, the row count nearly doubled, leading to major inconsistencies.

To fix this:

- We redefined **primary keys (indexes)** for every table.
- Used **Left Outer Joins** to ensure all flight records were preserved.
- Verified data accuracy after every merge using **reconciliation checks** (row count and total delay comparisons).

After several iterations, a clean, valid **star schema** was achieved as shown in the model diagram (Power Pivot screenshot).

4.5 Star Schema Structure

The final star schema consists of:

- **1 Fact Table:** Flight
- **6 Dimension Tables:** Date, Delay, Origin, Destination, Airline, Route

This design allows seamless connections between all aspects of flight operations while maintaining performance and consistency.

4.6 Primary Keys and Table Relationships

Table Name	Primary Key	Key Type	Description	Relationship
Flight (Fact)	*Flight_ID*	Surrogate (Auto Index)	Unique identifier for each flight record	Linked to all dimension keys (foreign keys)
Date (Dimension)	*Day_ID*	Surrogate	Unique identifier for each date record	1-to-Many with Flight.Day_ID
Delay (Dimension)	*Delay_ID*	Surrogate	Unique identifier for delay type record	1-to-Many with Flight.Delay_ID
Origin (Dimension)	*Origin_ID*	Surrogate	Unique identifier for each origin airport	1-to-Many with Flight.Origin_ID
Destination (Dimension)	*Destination_ID*	Surrogate	Unique identifier for each destination airport	1-to-Many with Flight.Destination_ID
Airline (Dimension)	*Airline_ID*	Surrogate	Unique identifier for each airline	1-to-Many with Flight.Airline_ID
Route (Dimension)	*Route_ID*	Surrogate	Unique identifier for each flight route	1-to-Many with Flight.Route_ID

4.7 Relationship Overview

Date → Flight

- **Type:** One-to-Many
- **Key:** Date.Day_ID = Flight.Day_ID
- **Purpose:** Enables time-based grouping (daily, monthly, weekday trends).

Delay → Flight

- **Type:** One-to-Many
- **Key:** Delay.Delay_ID = Flight.Delay_ID
- **Purpose:** Links each flight to its dominant delay cause for category-level insights.

Origin → Flight

- **Type:** One-to-Many
- **Key:** Origin.Origin_ID = Flight.Origin_ID
- **Purpose:** Connects flight operations with the departure airport's characteristics.

Destination → Flight

- **Type:** One-to-Many
- **Key:** Destination.Destination_ID = Flight.Destination_ID
- **Purpose:** Enables analysis of arrival performance by destination region or state.

Airline → Flight

- **Type:** One-to-Many
- **Key:** Airline.Airline_ID = Flight.Airline_ID
- **Purpose:** Allows performance evaluation by airline (e.g., delay averages, on-time %).

Route → Flight

- **Type:** One-to-Many
- **Key:** Route.Route_ID = Flight.Route_ID
- **Purpose:** Connects flight paths with distances, enabling analysis of route-level efficiency.

4.8 Importance of Modeling in Large Datasets

Proper data modeling brings several advantages:

- **Performance Optimization:** Queries and dashboards run faster since they access only relevant dimension data.
- **Data Integrity:** Relationships ensure consistency between airlines, airports, and flights.
- **Analytical Depth:** Enables insights like:
  - Top 10 airlines with the highest delay average
  - Delay causes by airport category
  - Delay duration trends by month or weekday
- **Scalability:** The structure can easily expand if new dimensions (e.g., Weather Conditions, Aircraft Type) are added later.

4.9 Verification and Quality Assurance

After building the model:

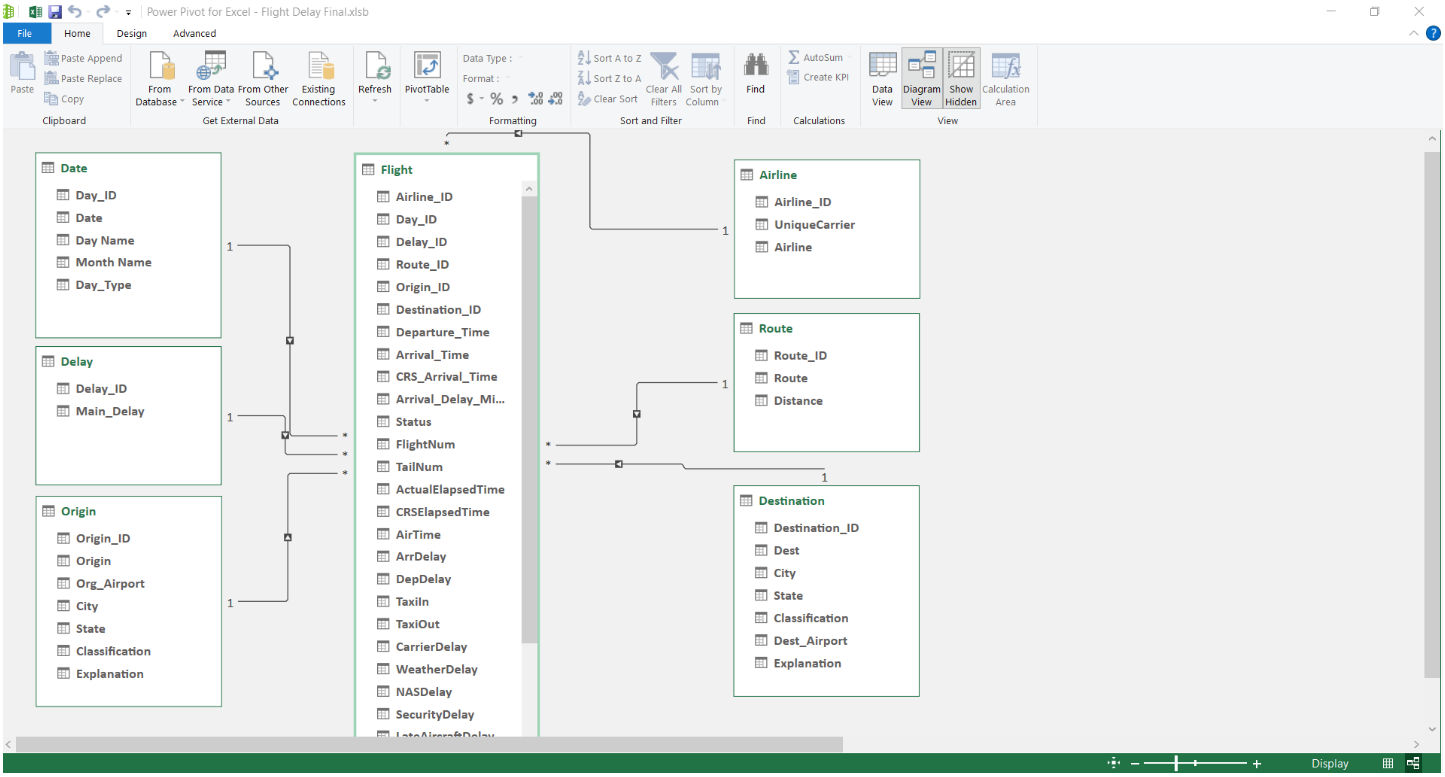
- **Row Counts** were compared before and after modeling to ensure no data loss.
- **TotalDelay** sums were verified for consistency.
- Random samples were checked to confirm correct relationship mapping.
- Power Pivot's **Diagram View** was used to visualize one-to-many relationships and confirm correct cardinality.

4.10 Modeling Outcome

The final data model represents a **fully functional star schema**, with:

- A **central fact table (Flight)** storing quantitative metrics.
- **Six connected dimension tables** providing descriptive attributes.
- All relationships established through **primary and foreign keys**, enabling dynamic filtering and drill-down across time, location, and delay causes.

This modeling process turned the dataset from a flat, redundant table into a structured, optimized model — ensuring reliable, flexible, and high-performance analysis in Power BI and Excel.


Power Pivot for Excel - Flight Delay Final.xlsx
File
Home
Design
Advanced
Date
Day_ID
Date
Day Name
Month Name
Day_Type
Delay
Delay_ID
Main_Delay
Origin
Origin_ID
Origin
Org_Airport
City
State
Classification
Explanation
Flight
Airline_ID
Day_ID
Delay_ID
Route_ID
Origin_ID
Destination_ID
Departure_Time
Arrival_Time
CRS_Arrival_Time
Arrival_Delay_Mi...
Status
FlightNum
TailNum
ActualElapsedTime
CRSElapsedTime
AirTime
ArrDelay
DepDelay
TaxiIn
TaxiOut
CarrierDelay
WeatherDelay
NASDelay
SecurityDelay
Airline
Airline_ID
UniqueCarrier
Airline
Route
Route_ID
Route
Distance
Destination
Destination_ID
Dest
City
State
Classification
Dest_Airport
Explanation
Display

3.5 Tools Used

The project made use of a diverse set of modern data analysis tools, each designed for a specific phase of the workflow — from data cleaning and transformation to modeling and interactive visualization.
These tools not only facilitated smooth execution of the project but also ensured precision, reproducibility, and professional presentation of results.

The following section provides an overview of each tool, its general background, typical applications in data analytics, and its specific role within this project.

1. Microsoft Excel (with Power Query)

Developer: Microsoft Corporation

General Overview:
Microsoft Excel is one of the oldest and most widely used spreadsheet applications in the world. First released in **1985**, Excel has evolved from a simple calculation grid into a powerful data analytics and business reporting platform.
It is primarily used for **data entry, organization, statistical analysis, and visualization**. Its core strengths lie in formulas, pivot tables, and data visualization capabilities.

In 2010, Microsoft introduced **Power Query** (originally known as Data Explorer), a powerful add-in that revolutionized Excel's capabilities for **data transformation and automation**. Power Query uses the **M Language** to reshape and clean data without manual effort, bridging the gap between Excel and advanced BI tools like Power BI.

Common Uses in Data Analytics:

- Data cleaning, transformation, and merging.
- Exploratory analysis using pivot tables and charts.
- Statistical modeling and financial forecasting.
- Automating repetitive data preparation workflows.

Key Advantages:

- User-friendly interface suitable for both beginners and professionals.
- Integrates easily with multiple data sources (CSV, databases, web).
- Power Query provides automation and transparency in transformations.

- Seamless integration with Power BI for business reporting.

Role in This Project:

- Served as the **primary tool** for data preprocessing and initial modeling.
- Power Query was used extensively for data cleaning and transformation — filling missing airport names, fixing date formats, and creating new analytical fields (*Main_Delay*, *TotalDelay*, *Route*, *Status*).
- The **first-level data model** was designed within Excel, including index keys and early star-schema separation.
- Developed **pivot tables** and a **preliminary dashboard** to visualize early insights before importing the data into Power BI.

Excel and Power Query together formed the **foundation** of the data preparation phase, turning the raw flight data into a structured, analysis-ready dataset.

2. MySQL Workbench

Developer: Oracle Corporation

General Overview:
MySQL Workbench is a flagship tool for working with the **MySQL relational database management system (RDBMS)**. Originally developed by **MySQL AB** and later acquired by **Oracle** in 2010, it is used globally for **database design, SQL development, and server administration**.

MySQL Workbench provides a visual interface for managing databases, designing schemas, and running SQL queries. It is widely used in data engineering, analytics, and application development to handle large structured datasets efficiently.

Common Uses in Data Analytics:

- Storing and managing large structured datasets.
- Writing SQL queries for aggregation, filtering, and joining tables.
- Building and testing data models for analytical pipelines.
- Performing data validation and KPI calculations.

Key Advantages:

- High performance for querying large datasets.
- Visual database modeling and ER diagram design.
- Supports SQL scripting, triggers, and stored procedures.

- Easy integration with BI tools such as Power BI and Tableau.

Role in This Project:

- The cleaned dataset was **uploaded into MySQL Workbench** for structured storage and query execution.
- Designed SQL scripts to calculate **key performance indicators (KPIs)** such as:
  - Average delay per airline and airport.
  - Delay percentage by cause and route.
  - Total number of delayed vs. on-time flights.
- Performed full analytical queries on every field to validate and cross-check results obtained from Excel and Power BI.

MySQL Workbench ensured data accuracy, consistency, and reliable performance metrics before moving into the visualization stage.

3. Microsoft Power BI

Developer: Microsoft Corporation

General Overview:
Power BI is a **business intelligence (BI)** platform developed by Microsoft, launched officially in **2015** as part of the Power Platform. It enables users to connect to various data sources, transform and model data, and build interactive dashboards for decision-making.

Power BI integrates seamlessly with Excel, SQL databases, and cloud platforms like Azure. It is commonly used in industries for **KPI monitoring, performance tracking, and predictive analytics**.

Common Uses in Data Analytics:

- Building interactive dashboards and data reports.
- Creating dynamic KPIs and visual storytelling.
- Integrating multiple data sources into a unified model.
- Performing DAX-based calculations for business metrics.

Key Advantages:

- Strong integration with Microsoft ecosystem (Excel, SQL, Azure).

- Intuitive drag-and-drop interface for visualization.
- DAX (Data Analysis Expressions) language for custom measures.
- Cloud and desktop publishing capabilities.

Role in This Project:

- Served as the **main analytical and visualization tool**.
- The processed and modeled data from Excel and MySQL was imported into Power BI.
- Created multiple **DAX measures** to calculate metrics such as:
    - Average delay per airline.
    - Delay distribution by cause.
    - On-time performance ratios.
    - Delay trends across days, months, and airports.
- Developed a **comprehensive dashboard** that included charts, maps, and KPIs summarizing the performance of the U.S. flight network.

Power BI transformed the prepared dataset into an interactive, insight-rich analytical product — ready for stakeholders and presentations.

4. Tableau

Developer: Salesforce, Inc.

General Overview:
Tableau was founded in **2003** and has grown into one of the world's most respected data visualization platforms. Acquired by **Salesforce** in **2019**, Tableau is known for its fast, interactive analytics and ability to handle large-scale visual exploration with minimal technical setup.

It specializes in **data storytelling** through drag-and-drop visual design and strong support for geographic and temporal visualizations. Tableau is often used by analysts, researchers, and business professionals for **exploratory data analysis (EDA)** and executive dashboards.

Common Uses in Data Analytics:

- Visualizing large, complex datasets in real time.
- Creating geographical maps and trend analysis.
- Comparing KPIs across multiple categories.
- Presenting analytical findings in a business-friendly format.

Key Advantages:

- Fast rendering of visuals even with large datasets.
- Drag-and-drop simplicity with deep analytical power.
- Native connection to databases like MySQL, Excel, and cloud platforms.
- Strong storytelling and dashboard sharing capabilities.

Role in This Project:

- Used to design **comparative dashboards** and validate Power BI results.
- Created visual maps and charts showing delay frequency across routes and airports.
- Supported trend analysis of delay causes and time patterns with an emphasis on visual clarity and interactivity.

Tableau added an additional visual perspective to the project, enhancing storytelling and reinforcing analytical conclusions derived from Power BI.

Summary

Each tool played a vital and complementary role throughout the project's workflow:

- **Excel & Power Query:** Data preparation, cleaning, transformation, and preliminary visualization.
- **MySQL Workbench:** Structured data management, KPI calculation, and analytical validation.
- **Power BI:** Advanced data modeling, DAX-driven analysis, and interactive dashboards.
- **Tableau:** Enhanced visual comparison and storytelling.

Together, these tools created an end-to-end data analysis environment — from raw flight data to professional insights and performance dashboards.

3.6 Method

6.1 Introduction

This chapter provides a comprehensive explanation of the analytical workflow followed in this project — from raw data ingestion and preprocessing to data modeling, SQL analysis, and visualization.

The project was executed using three main environments: Microsoft Excel (with Power Query) for preprocessing, MySQL Workbench for relational querying and validation, and Microsoft Power BI for analytical modeling and interactive visualization.

Every stage is documented with detailed methodology, examples of formulas and scripts, and clear justifications for each decision to ensure reproducibility, transparency, and data integrity.

6.2 Environment and Project Structure

6.2.1 File Organization

The project was structured into organized directories to ensure a clear workflow and separation of responsibilities:

```
/FinalProject
  /Data
    Flight_Delay_Raw.csv
    Flight_Delay_Cleaned.xlsx
  /Excel
    Flight Delay Final.xlsb      --> Power Query, Pivot Tables, Preliminary Dashboard
  /SQL
    create_tables.sql
    load_data.sql
    queries_kpis.sql
  /PowerBI
```

FlightDelay.pbix

/Docs

Documentation.docx

6.2.2 Workflow Overview

Load and explore the raw data in Excel.

Perform data cleaning and preprocessing using Power Query.

Export the cleaned dataset into MySQL for SQL-based validation and KPI generation.

Import validated tables into Power BI for modeling and DAX-driven analysis.

Build final dashboards integrating all insights with Power BI.

Build final dashboards integrating all insights with Tableau.

6.3 Data Understanding and Initial Exploration (EDA)

6.3.1 Dataset Summary

The dataset titled "Flight Delay and Causes" contains information about U.S. domestic flights between January 1, 2019, and June 30, 2019, totaling 484,552 rows and 28 columns.

Columns include flight details (Airline, Origin, Destination, Date, DepDelay, ArrDelay), causes of delay, and distance-related information.

6.3.2 EDA Procedures

Using Excel and SQL, an exploratory analysis was conducted to:

Calculate descriptive statistics (mean, median, min, max, std deviation).

Identify missing values across columns.

Detect anomalies or outliers in delay durations and time fields.

Understand distribution of airlines, routes, and delay causes.

This stage confirmed that the dataset was large, rich, but required substantial preprocessing before modeling.

6.4 Data Cleaning

6.4.1 Identified Issues

Missing airport names in Org_Airport and Dest_Airport.

Incorrect date formats (month/day swapped with day/month).

Time columns (DepTime, ArrTime, CRSArrTime) stored as integer values.

Presence of cancelled or diverted flights that could distort delay metrics.

	FlightNum	TailNum	ActualElapsedTime	CRSElapsedTime	AirTime	ArrDelay	DepDelay	Origin	Org_Airport	Dest	Dest_Airport	Distance	TaxiIn
17168	5741	N218SW	33	37	20	40	44	MOD	#N/A	SFO	San Francisco In	78	5
17171	5761	N582SW	74	85	63	42	53	MOD	#N/A	LAX	Los Angeles Inte	292	7
17172	5769	N295SW	84	90	61	77	83	MOD	#N/A	LAX	Los Angeles Inte	292	10
17624	5769	N584SW	93	90	67	94	91	MOD	#N/A	LAX	Los Angeles Inte	292	17
18051	5720	N566SW	43	39	32	18	14	MOD	#N/A	SFO	San Francisco In	78	4
18054	5737	N564SW	72	61	56	162	151	CIC	#N/A	SFO	San Francisco In	153	6
18058	5756	N218SW	91	59	71	154	122	CIC	#N/A	SFO	San Francisco In	153	7
18066	5790	N218SW	76	59	55	205	188	CIC	#N/A	SFO	San Francisco In	153	5
18179	6456	N965SW	78	76	63	153	151	PMD	#N/A	SFO	San Francisco In	316	6
18320	5430	N288SW	193	66	61	221	94	IYK	#N/A	LAX	Los Angeles Inte	123	105
18344	5458	N582SW	45	39	16	63	57	OXR	#N/A	LAX	Los Angeles Inte	49	15
18430	5724	N224SW	35	38	24	156	159	MOD	#N/A	SFO	San Francisco In	78	4
18437	5740	N295SW	81	61	62	110	90	IPL	#N/A	LAX	Los Angeles Inte	181	10
18440	5741	N218SW	44	37	27	314	307	MOD	#N/A	SFO	San Francisco In	78	7
18448	5756	N562SW	65	59	56	109	103	CIC	#N/A	SFO	San Francisco In	153	6
18452	5761	N296SW	87	85	71	47	45	MOD	#N/A	LAX	Los Angeles Inte	292	11
18455	5769	N296SW	95	90	73	22	17	MOD	#N/A	LAX	Los Angeles Inte	292	16
18458	5775	N227SW	27	31	14	89	93	IPL	#N/A	YUM	Yuma Internatio	58	5
18460	5778	N393SW	28	31	15	17	20	IPL	#N/A	YUM	Yuma Internatio	58	4
18550	6456	N946SW	115	76	60	203	164	PMD	#N/A	SFO	San Francisco In	316	11
18922	5458	N292SW	49	39	26	31	21	OXR	#N/A	LAX	Los Angeles Inte	49	7
18959	5516	N229SW	62	52	38	77	67	IYK	#N/A	LAX	Los Angeles Inte	123	17
19033	5720	N270YV	51	39	32	18	6	MOD	#N/A	SFO	San Francisco In	78	6
19035	5724	N569SW	49	38	31	209	198	MOD	#N/A	SFO	San Francisco In	78	8
19042	5737	N216SW	63	61	51	252	250	CIC	#N/A	SFO	San Francisco In	153	4
19045	5741	N221SW	41	37	29	371	367	MOD	#N/A	SFO	San Francisco In	78	5
19052	5751	N295UX	89	86	64	148	145	MOD	#N/A	LAX	Los Angeles Inte	292	15
19055	5756	N295SW	64	59	56	153	148	CIC	#N/A	SFO	San Francisco In	153	4
19059	5759	N292SW	100	92	63	73	65	MOD	#N/A	LAX	Los Angeles Inte	292	25
19061	5761	N250YV	77	85	62	77	85	MOD	#N/A	LAX	Los Angeles Inte	292	7
19072	5775	N250YV	32	31	16	50	49	IPL	#N/A	YUM	Yuma Internatio	58	2
19258	6456	N970SW	96	76	70	195	175	PMD	#N/A	SFO	San Francisco In	316	5

Flight_delay

	DayOfWeek	Date	DepTime	ArrTime	CRSArrTime	UniqueCarrier
2	4	3/1/2019	1829	1959	1925	WN
3	4	3/1/2019	1937	2037	1940	WN
4	4	3/1/2019	1644	1845	1725	WN
5	4	3/1/2019	1452	1640	1625	WN
6	4	3/1/2019	1323	1526	1510	WN
7	4	3/1/2019	1416	1512	1435	WN
8	4	3/1/2019	1657	1754	1735	WN
9	4	3/1/2019	1422	1657	1610	WN
10	4	3/1/2019	2107	2334	2230	WN
11	4	3/1/2019	1812	1927	1815	WN
12	4	3/1/2019	1326	1559	1530	WN
13	4	3/1/2019	1450	1806	1745	WN
14	4	3/1/2019	2245	2354	1850	WN
15	4	3/1/2019	2025	2135	2100	WN
16	4	3/1/2019	1038	1314	1225	WN
17	4	3/1/2019	1900	2123	2045	WN
18	4	3/1/2019	948	959	940	WN
19	4	3/1/2019	646	725	655	WN
20	4	3/1/2019	1110	1136	1110	WN
21	4	3/1/2019	1813	1936	1905	WN
22	4	3/1/2019	1734	1941	1905	WN
23	4	3/1/2019	958	1052	950	WN
24	4	3/1/2019	1538	1753	1710	WN
25	4	3/1/2019	2248	102	2345	WN
26	4	3/1/2019	1327	1550	1500	WN
27	4	3/1/2019	1832	148	30	WN
28	4	3/1/2019	1229	1633	1555	WN
29	4	3/1/2019	2118	144	50	WN
30	4	3/1/2019	1739	114	25	WN
31	4	3/1/2019	1325	1841	1810	WN
32	4	3/1/2019	1506	2030	2010	WN
33	4	3/1/2019	2039	155	55	WN

Flight_delay

6.4.2 Cleaning Actions

A. Filling Missing Airport Names

Used Origin and Dest IATA codes as reference keys.

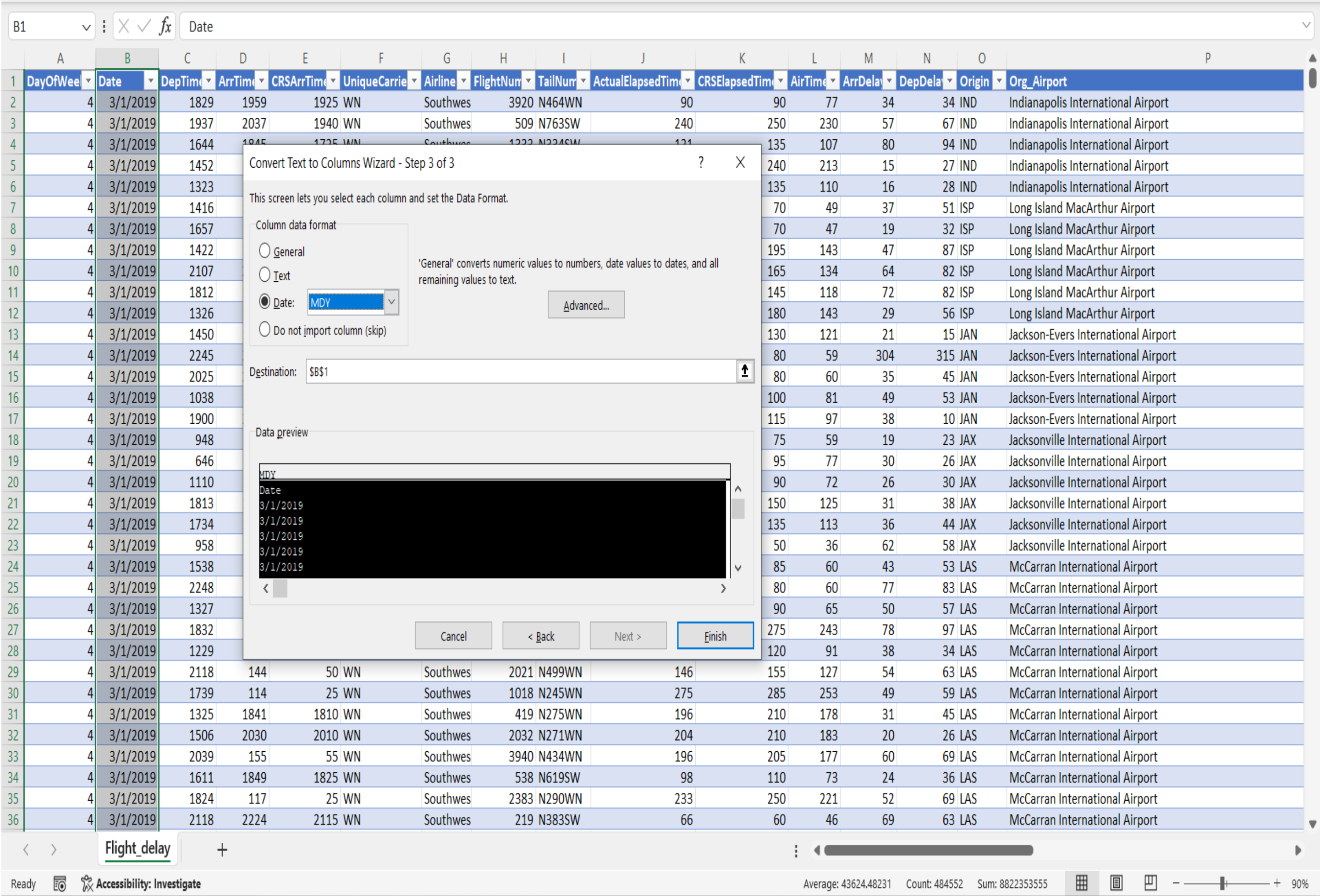
Searched official airport directories using automation and manual lookups (assisted by ChatGPT queries).

Replaced null values in Org_Airport and Dest_Airport with accurate full airport names.

B. Fixing Date Formats

In Excel, used Text to Columns to split date components and reorder them to DD/MM/YYYY.

In Power Query, used culture-aware date parsing functions for consistent format conversion.

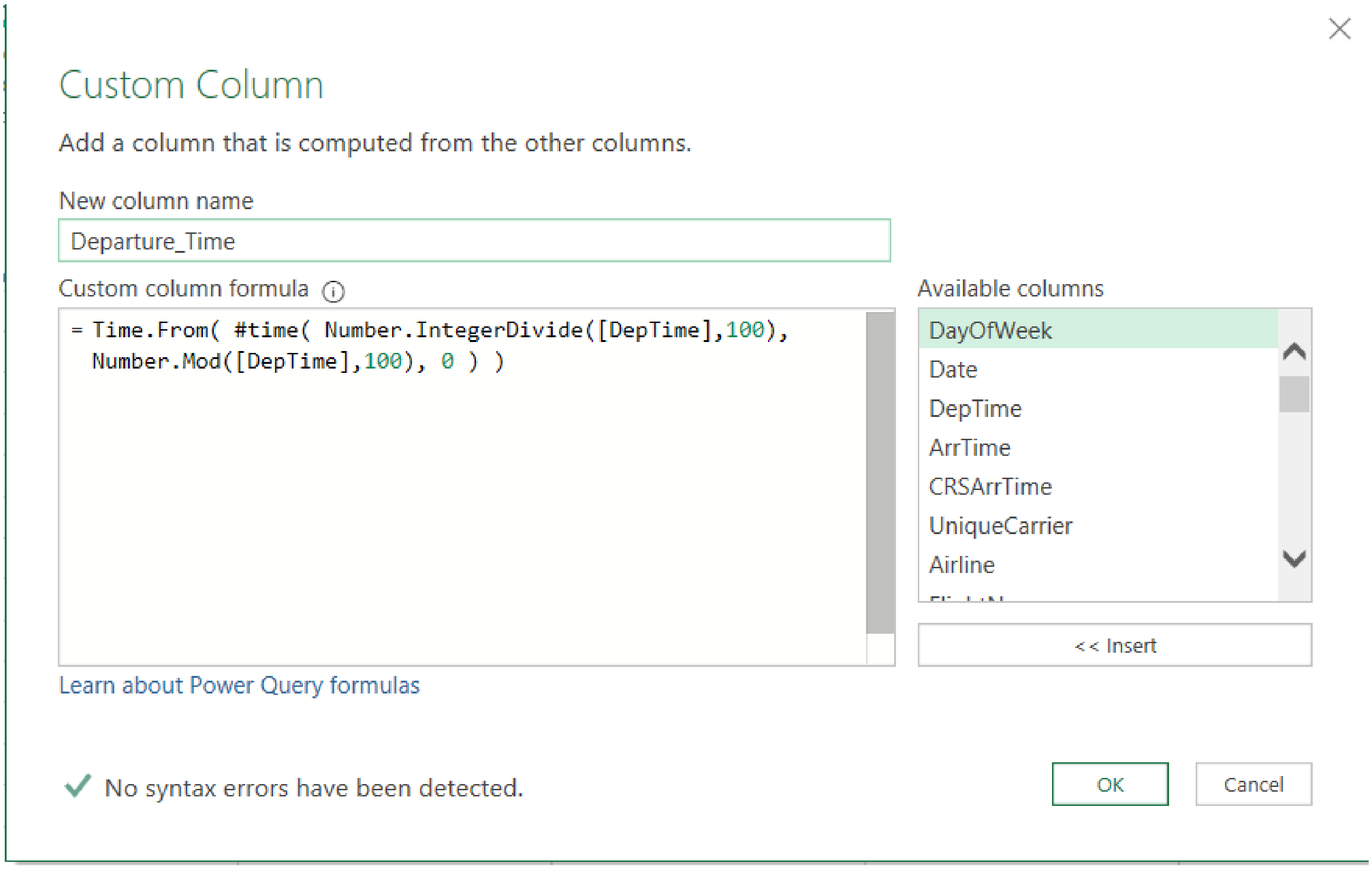


C. Converting Time Columns

Used Power Query M expressions to convert integer values into proper Time format:

Time.From(#time(Number.IntegerDivide([DepTime],100), Number.Mod([DepTime],100), 0))

Applied the same transformation to ArrTime and CRSArrTime.



D. Replacing Null Delay Values

All nulls in delay causes were replaced with zeros using Power Query:

if [CarrierDelay] = null then 0 else [CarrierDelay]

This ensured accurate calculations of total delays without bias.

E. Excluding Cancelled/Diverted Flights

Rows with Cancelled = 1 or Diverted = 1 were filtered out from the analytical dataset, as their delay data is not comparable to operational flights.

6.5 Preprocessing and Feature Engineering

6.5.1 Objective

To enrich the dataset and enhance analytical granularity by deriving new calculated fields, standardizing formats, and creating index keys for modeling.

6.5.2 New Columns Added

Column	Description	Example / Logic
TotalDelay	Sum of all delay causes	CarrierDelay + WeatherDelay + NASDelay + SecurityDelay + LateAircraftDelay
Main_Delay	Dominant cause of delay	Power Query M code comparing max among delay types
Delay_Ratios	Percentage of each cause vs TotalDelay	(CarrierDelay / TotalDelay) * 100
Status	Categorical indicator (Early, On Time, Delayed)	If ArrDelay > 15 → “Delayed”
Route	Combined Origin–Destination code	Origin & "-" & Dest
DayName / MonthName	Derived from Date field	Power Query Date.DayOfWeekName and Date.MonthName
Origin_City / State	Enriched via airport lookup	Matched from IATA registry
Dest_City / State	Same as above	-
Airport_Classification	Large/Medium/Small Hub categorization	Based on airport passenger traffic
Index Columns	Surrogate keys for each dimension	Auto-generated by Power Query AddIndexColumn

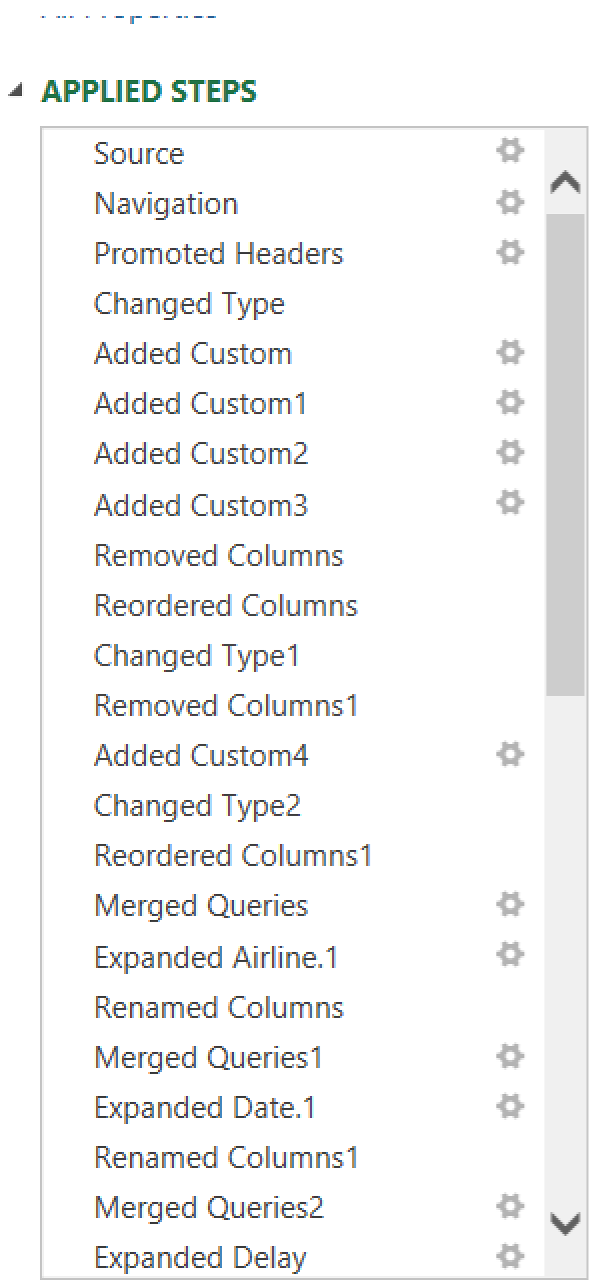
6.5.3 Purpose of Transformations

Simplify visual analysis (Status, Route).

Enable aggregation by geography and time (City, State, Month, Day).

Support dimension linking in star schema (Index columns).

Ensure data integrity for KPI calculations.



6.6 Data Modeling (Star Schema Design)

6.6.1 Why Data Modeling Was Necessary

Since the dataset contained almost half a million records and repeated information (airlines, airports, routes), it was inefficient to analyze as a flat file.

Therefore, a data warehouse approach was adopted using a Star Schema design to:

Reduce redundancy

Optimize performance

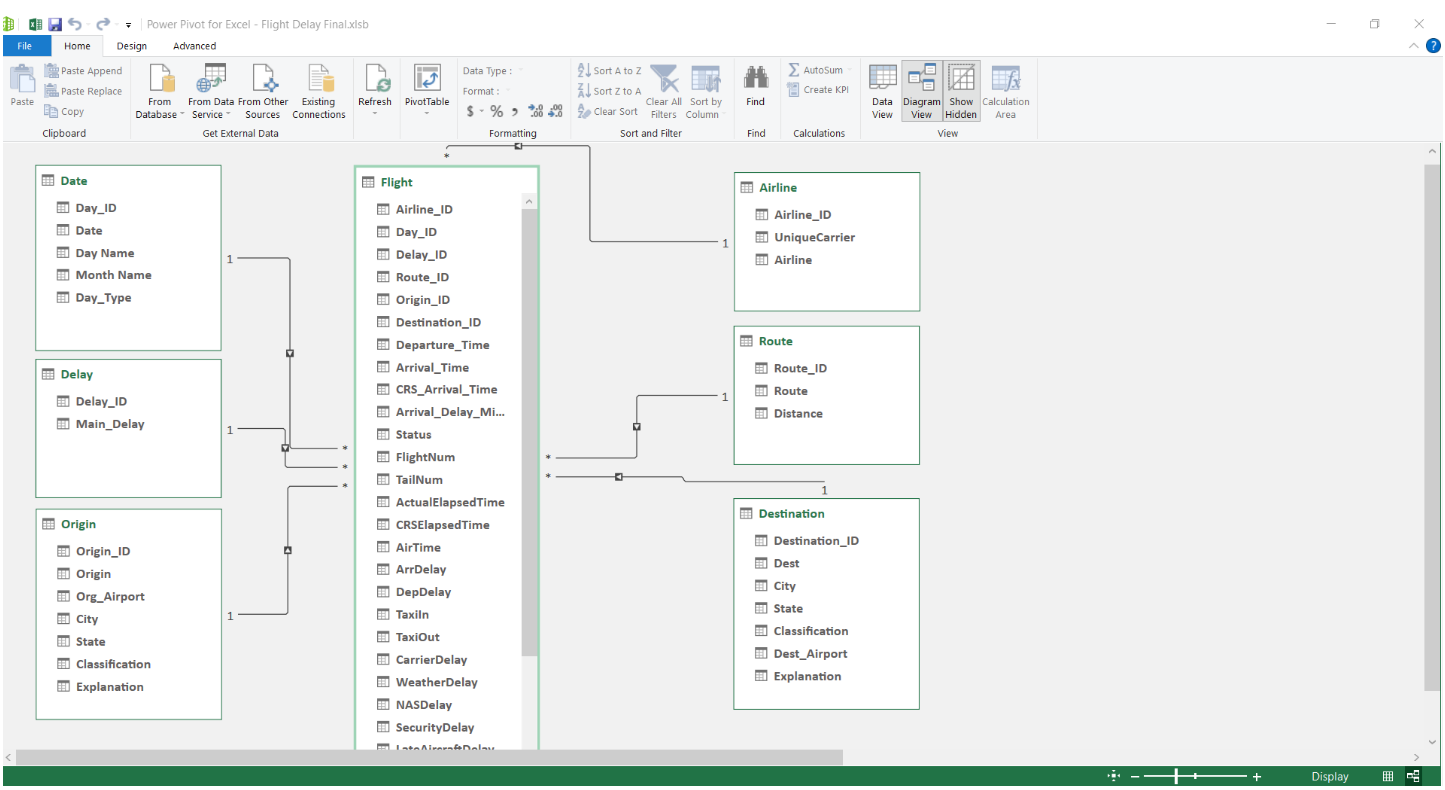
Enable flexible relational analysis

6.6.2 Model Structure

Fact Table: Flight (contains numeric metrics and foreign keys).

Dimension Tables: Date, Airline, Origin, Destination, Route, and Delay.

Each dimension has a unique Primary Key (ID) generated via indexing, while the fact table references them as Foreign Keys.



6.6.3 Implementation Steps

Create distinct tables in Power Query for each dimension.

Add Index Columns (Airline_ID, Origin_ID, Dest_ID, Route_ID, Day_ID, Delay_ID).

Merge tables using Left Joins to insert Foreign Keys into Flight.

Validate record counts and ensure no data loss.

Export finalized tables for SQL loading and BI modeling.

6.6.4 Validation Checks

Row Count Check: Ensure number of records in Flight unchanged.

TotalDelay Sum Check: Pre vs post modeling values match.

Distinct Count Check: Validate unique keys per dimension.

6.7 SQL Implementation and KPI Generation

6.7.1 Purpose of Using MySQL

To validate the relational design and extract KPI summaries efficiently. SQL allows high performance and structured querying of large datasets.

6.7.2 Core SQL Scripts

Table Creation:

```
CREATE TABLE flight (
  Flight_ID INT PRIMARY KEY,
  FlightDate DATE,
  Airline_ID INT,
  Origin_ID INT,
  Dest_ID INT,
  TotalDelay INT,
  CarrierDelay INT,
  WeatherDelay INT,
  NASDelay INT,
  SecurityDelay INT,
  LateAircraftDelay INT
);
```

```sql
SET GLOBAL local_infile = 1;
CREATE DATABASE IF NOT EXISTS new;
USE new;

-- create table

CREATE TABLE flight (
    DayOfWeek int,
    date VARCHAR(20),
    DepTime time,
    ArrTime time,
    CRSArrTime time,
    UniqueCarrier varchar(5),
    Airline varchar(100),
    FlightNum varchar(10),
    TailNum varchar(10),
    ActualElapsedTime int,
    CRSElapsedTime INT,
```

Data Loading:

LOAD DATA INFILE '/path/Flight_Cleaned.csv'

INTO TABLE flight

FIELDS TERMINATED BY ','

ENCLOSED BY '"'

IGNORE 1 LINES;

```sql
-- insert data into the table
LOAD DATA LOCAL INFILE 'E:\\desktop\\projecr\\Flight Delay My sql.csv'
INTO TABLE flight
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
select * from flight;
```

Example KPI Queries:

Average Delay per Airline

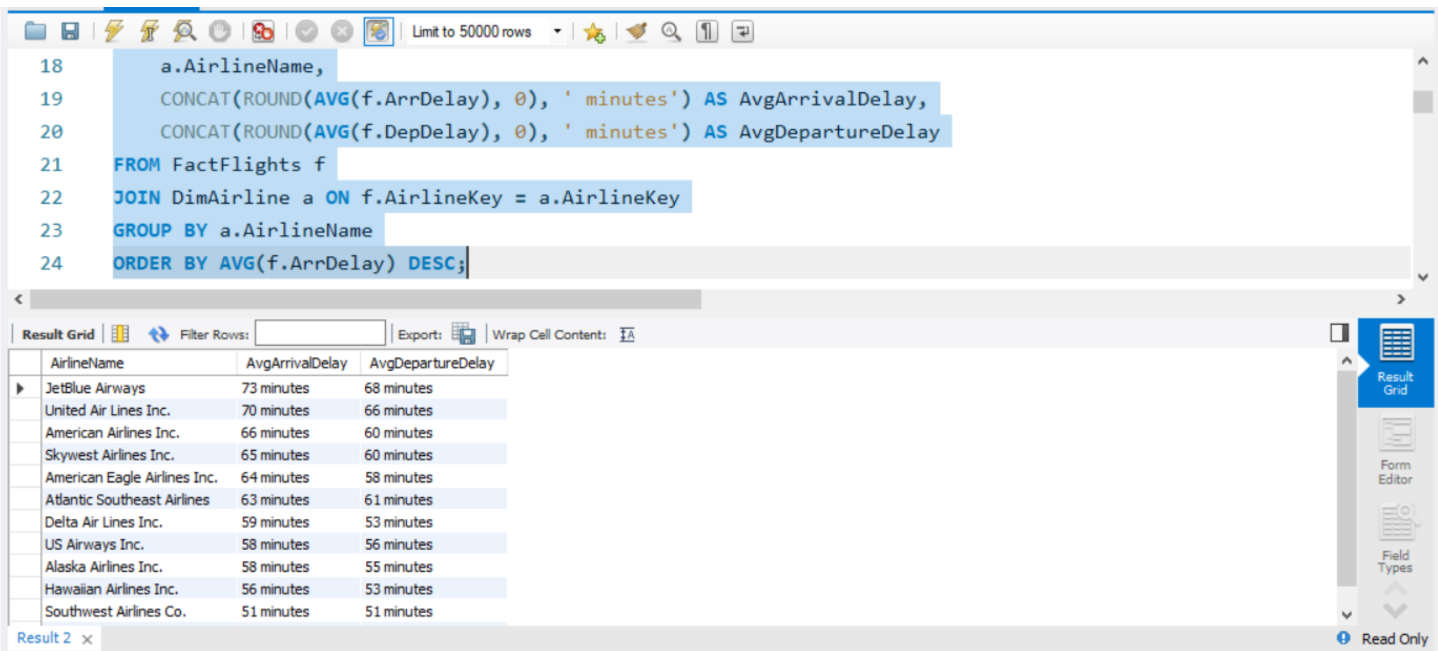
SELECT a.Airline, ROUND(AVG(f.TotalDelay),2) AS AvgDelay, COUNT(*) AS Flights

FROM flight f

JOIN airline a ON f.Airline_ID = a.Airline_ID

GROUP BY a.Airline

ORDER BY AvgDelay DESC;



Delay Breakdown by Cause

SELECT a.Airline,

SUM(f.CarrierDelay) AS Carrier,

SUM(f.WeatherDelay) AS Weather,

SUM(f.NASDelay) AS NAS,

SUM(f.SecurityDelay) AS Security,

SUM(f.LateAircraftDelay) AS LateAircraft

FROM flight f

JOIN airline a ON f.Airline_ID = a.Airline_ID

GROUP BY a.Airline;

Monthly Trend Analysis

SELECT DATE_FORMAT(FlightDate, '%Y-%m') AS Month, AVG(TotalDelay) AS AvgMonthlyDelay

FROM flight

GROUP BY Month

ORDER BY Month;

6.8 Excel Pivot Tables and Preliminary Dashboard

6.8.1 Objective

Before migrating to BI tools, a preliminary dashboard was developed in Excel using Pivot Tables to quickly validate patterns and insights.

6.8.2 Implementation

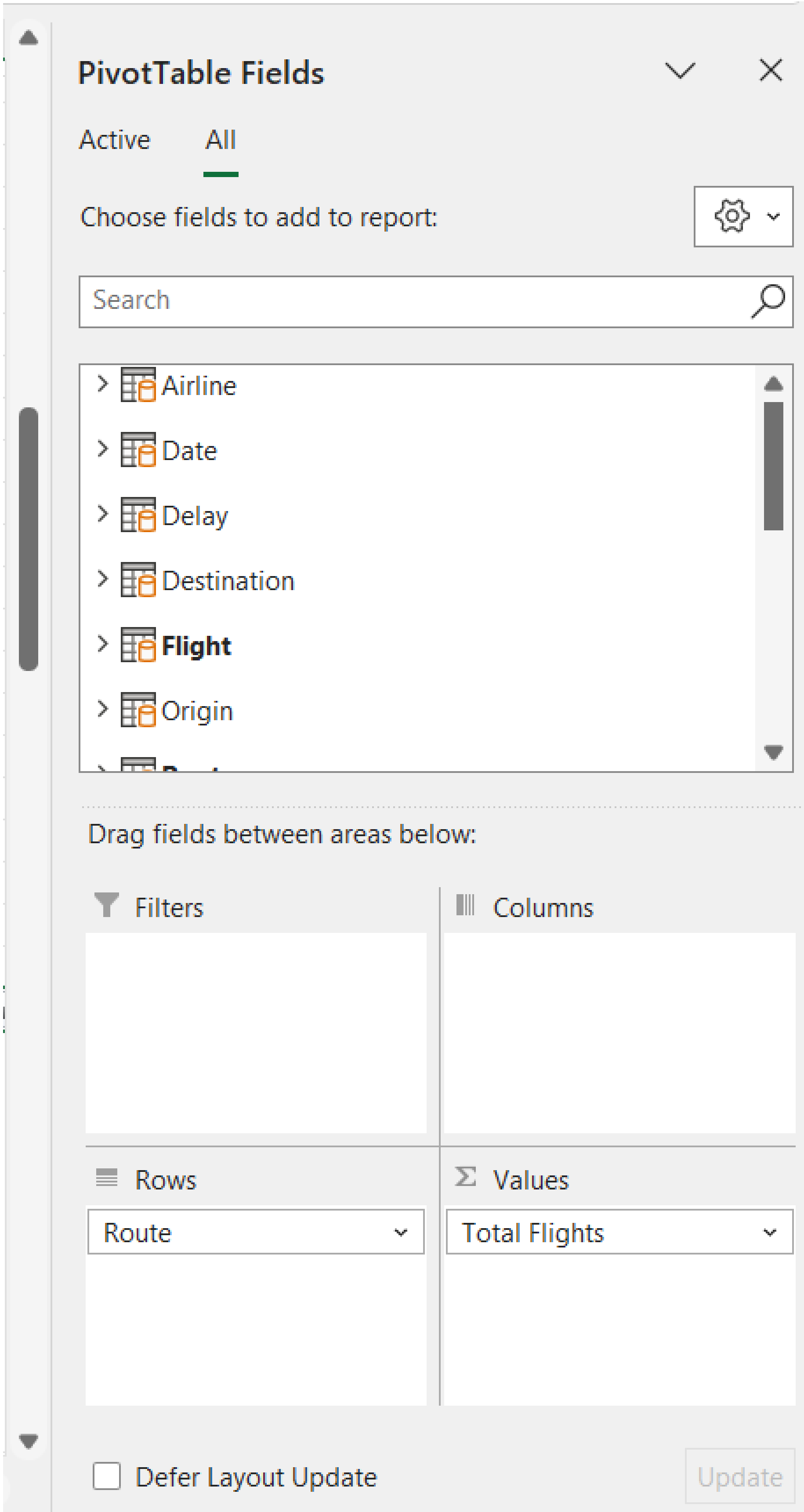
Pivot Data Source: Cleaned Flight table.

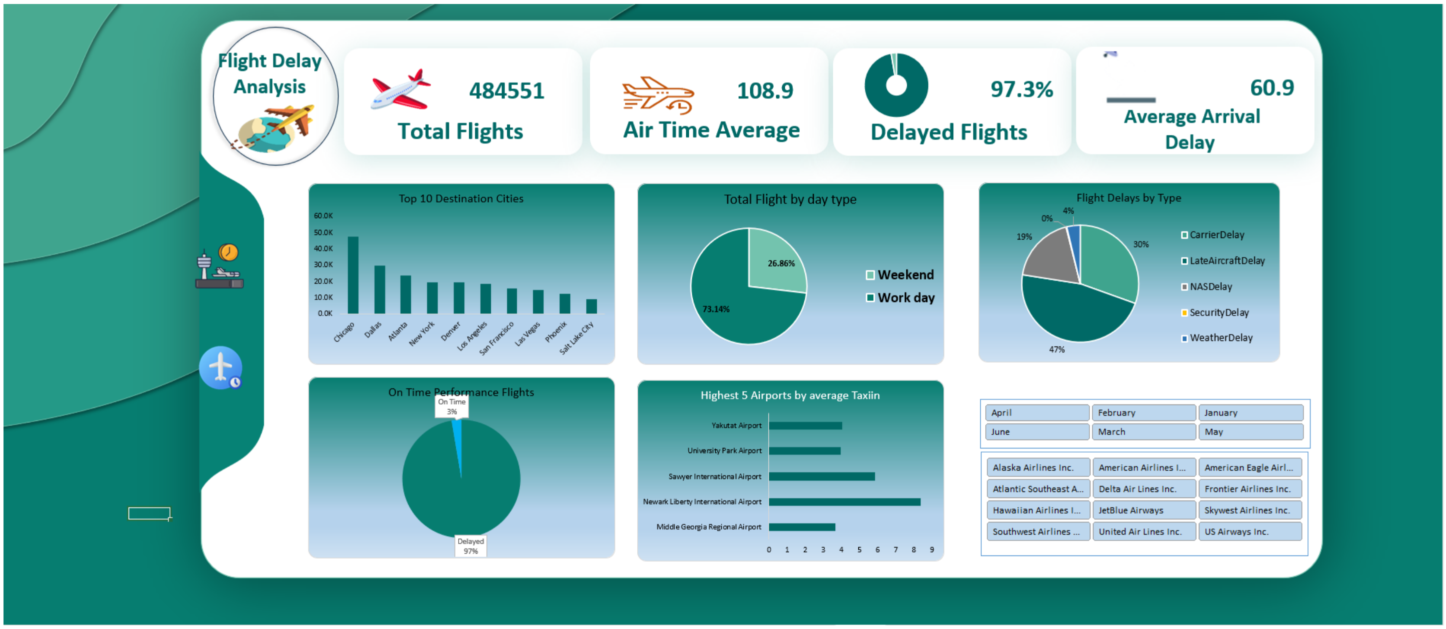
Values: COUNT(Flight_ID), SUM(TotalDelay), AVG(ArrDelay).

Rows/Columns: Airline, Route, MonthName, DayName.

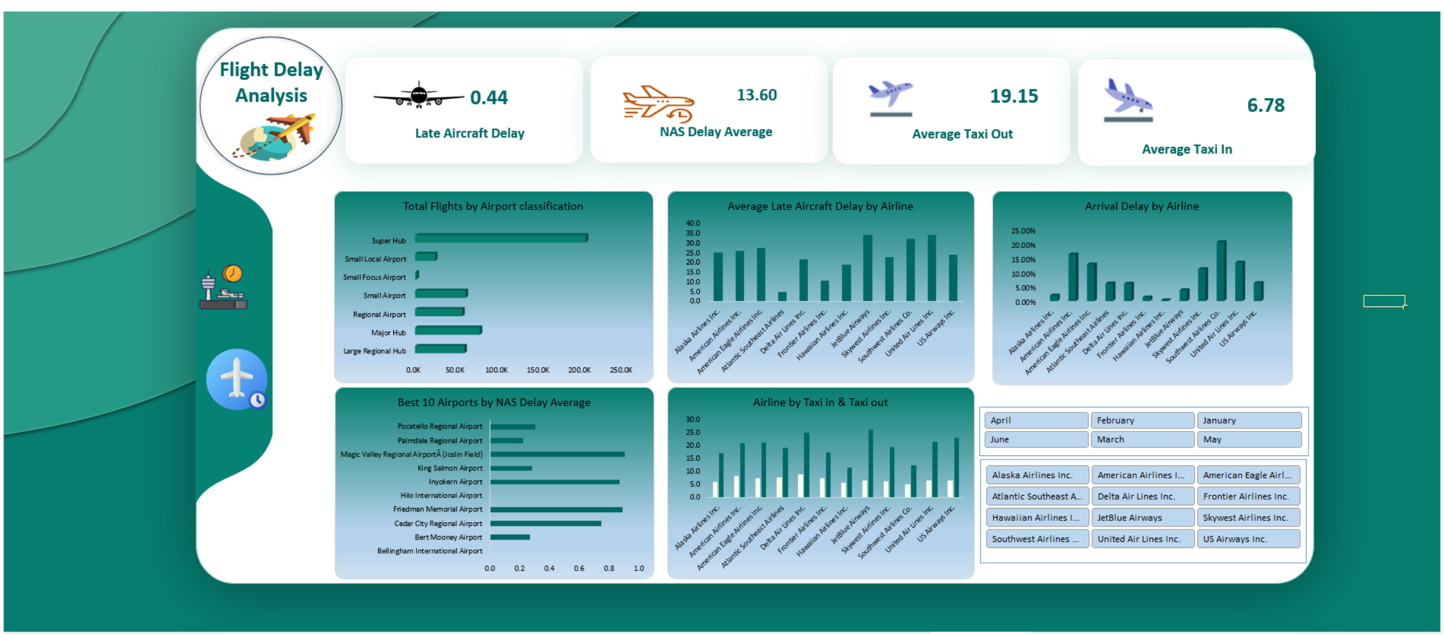
Slicers: Status, Main_Delay, Airport Classification.

Charts: Bar, Line, and Pie visualizations for basic insight checking.


PivotTable Fields
Active
All
Choose fields to add to report:
Search
Airline
Date
Delay
Destination
Flight
Origin
Drag fields between areas below:
Filters
Columns
Rows
Values
Route
Total Flights
Defer Layout Update
Update


Flight Delay Analysis
484551
Total Flights
108.9
Air Time Average
97.3%
Delayed Flights
60.9
Average Arrival Delay
Top 10 Destination Cities
Chicago
Dallas
Atlanta
New York
Denver
Los Angeles
San Francisco
Las Vegas
Phoenix
Salt Lake City
Total Flight by day type
26.86%
73.14%
Weekend
Work day
Flight Delays by Type
CarrierDelay
LateAircraftDelay
NASDelay
SecurityDelay
WeatherDelay
On Time Performance Flights
On Time 3%
Delayed 97%
Highest 5 Airports by average Taxiin
Yakutat Airport
University Park Airport
Sawyer International Airport
Newark Liberty International Airport
Middle Georgia Regional Airport
April
February
January
June
March
May
Alaska Airlines Inc.
Atlantic Southeast A...
Hawaiian Airlines I...
Southwest Airlines ...
Delta Air Lines Inc.
JetBlue Airways
United Air Lines Inc.
Frontier Airlines Inc.
Skywest Airlines Inc.
US Airways Inc.


Flight Delay Analysis
3.15
Weather Delay Average
17.42
Carrier Delay Average
0.08
Security Delay Average
60.9
Average Arrival Delay
5 Highest Destinations in weather delay
California
Florida
Georgia
Illinois (O'Hare)
Fort Worth
Airline has the highest Carrier delay
Average Security Delay by Airline
Which month has the highest weather delayes?
February
June
January
March
April
May
Alaska Airlines Inc.
American Airlines Inc.
American Eagle Airlines Inc.
Atlantic Southeast Airlines
Delta Air Lines Inc.
Frontier Airlines Inc.
Hawaiian Airlines Inc.
JetBlue Airways
Skywest Airlines Inc.
Southwest Airlines Co.
United Air Lines Inc.
US Airways Inc.


Flight Delay Analysis
0.44
Late Aircraft Delay
13.60
NAS Delay Average
19.15
Average Taxi Out
6.78
Average Taxi In
Total Flights by Airport classification
Super Hub
Small Local Airport
Small Focus Airport
Small Airport
Regional Airport
Major Hub
Large Regional Hub
Average Late Aircraft Delay by Airline
Arrival Delay by Airline
Best 10 Airports by NAS Delay Average
Pocatello Regional Airport
Palmdale Regional Airport
King Salmon Airport
Inyokern Airport
Hilo International Airport
Friedman Memorial Airport
Cedar City Regional Airport
Bert Mooney Airport
Bellingham International Airport
Airline by Taxi in & Taxi out
April
February
January
June
March
May

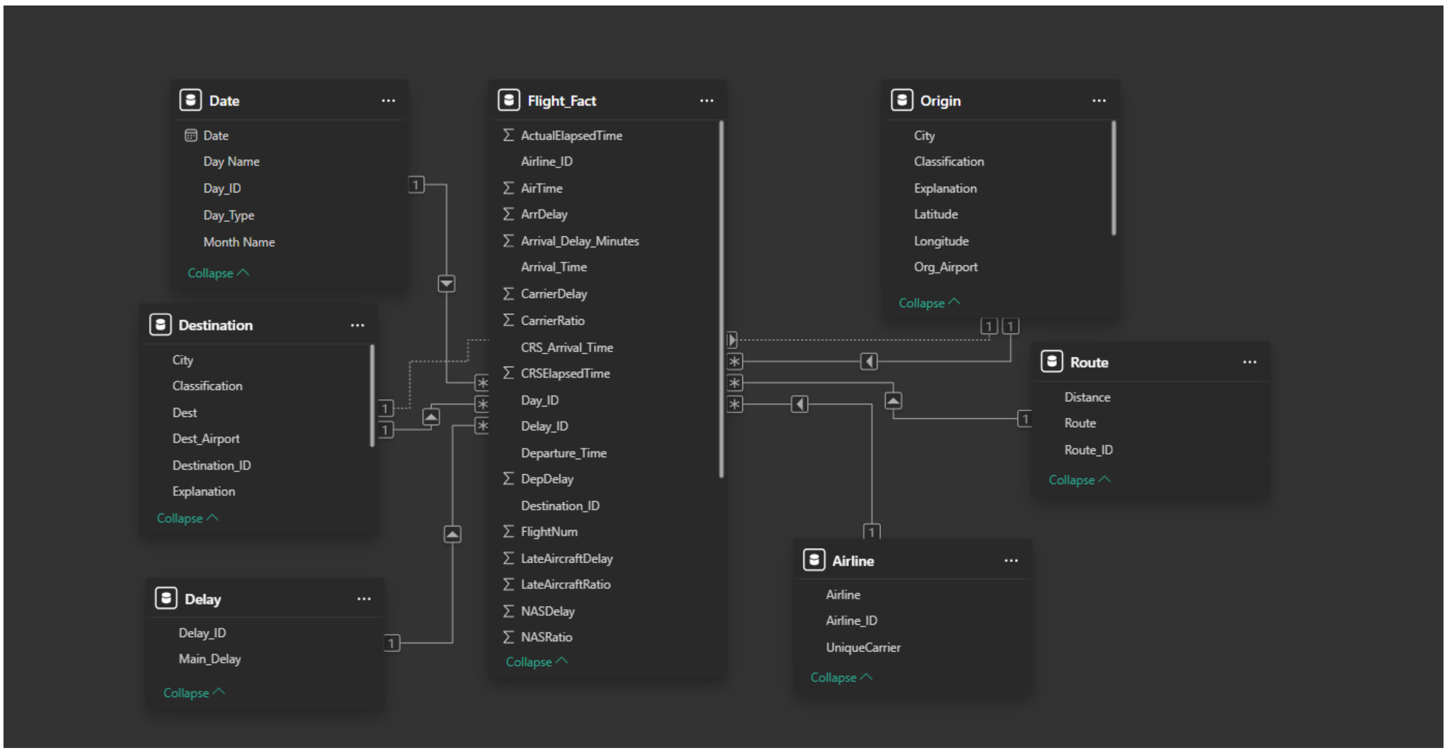
6.9 Power BI Modeling and Visualization

6.9.1 Data Import and Relationships

Imported the same star schema into Power BI.

Defined One-to-Many relationships between dimensions and fact.

Verified directional filtering and relationship cardinality.



6.9.2 DAX Measures Examples

Total Flights = COUNTROWS(Flight)

Total Delay Minutes = SUM(Flight[TotalDelay])

Avg Delay per Flight = DIVIDE([Total Delay Minutes], [Total Flights])

OnTime % = DIVIDE(

CALCULATE(COUNTROWS(Flight), Flight[Status] = "On Time"),

[Total Flights]

)

LateAircraft % = DIVIDE(SUM(Flight[LateAircraftDelay]), [Total Delay Minutes])

6.9.3 Visual Components

KPIs: Total Flights, Total Delayed Flights, Average Delay, On-Time %.

Trend Chart: Average delay by month.

Bar Chart: Average delay by airline.

Map Visualization: Delay by airport location.

Pie/Stacked Bar: Delay cause distribution.

Table: Top 10 delayed routes.

Slicers: Airline, Month, DayName, Airport Classification.

6.10 Validation and Testing

To ensure the reliability of the entire pipeline:

Record Consistency: Compared row counts across Excel, MySQL, and Power BI.

Value Validation: Matched total delay sums across all tools.

Sample Verification: Checked random flight IDs for consistency.

Cross-Tool Comparison: Confirmed identical outputs from SQL queries and DAX calculations.

Edge Case Testing: Ensured correct handling of early flights and zero-delay records.

6.11 Final Dashboard Design

The final Power BI dashboard integrates all KPIs and visuals in one interactive interface:

Header: Project title, date range (Jan–Jun 2019).

KPI Cards: Total Flights, Total Delay Minutes, On-Time %.

Filters: Airline, Month, Day, Delay Cause, Airport Type.

Visual Area:

Line chart: Monthly trend.

Bar chart: Delay by airline.

Map: Geographic delay overview.

Pie chart: Delay causes.

Table: Top routes.

Color palette: Neutral blues and oranges for clarity.

Enabled drill-down and tooltips for interactivity.

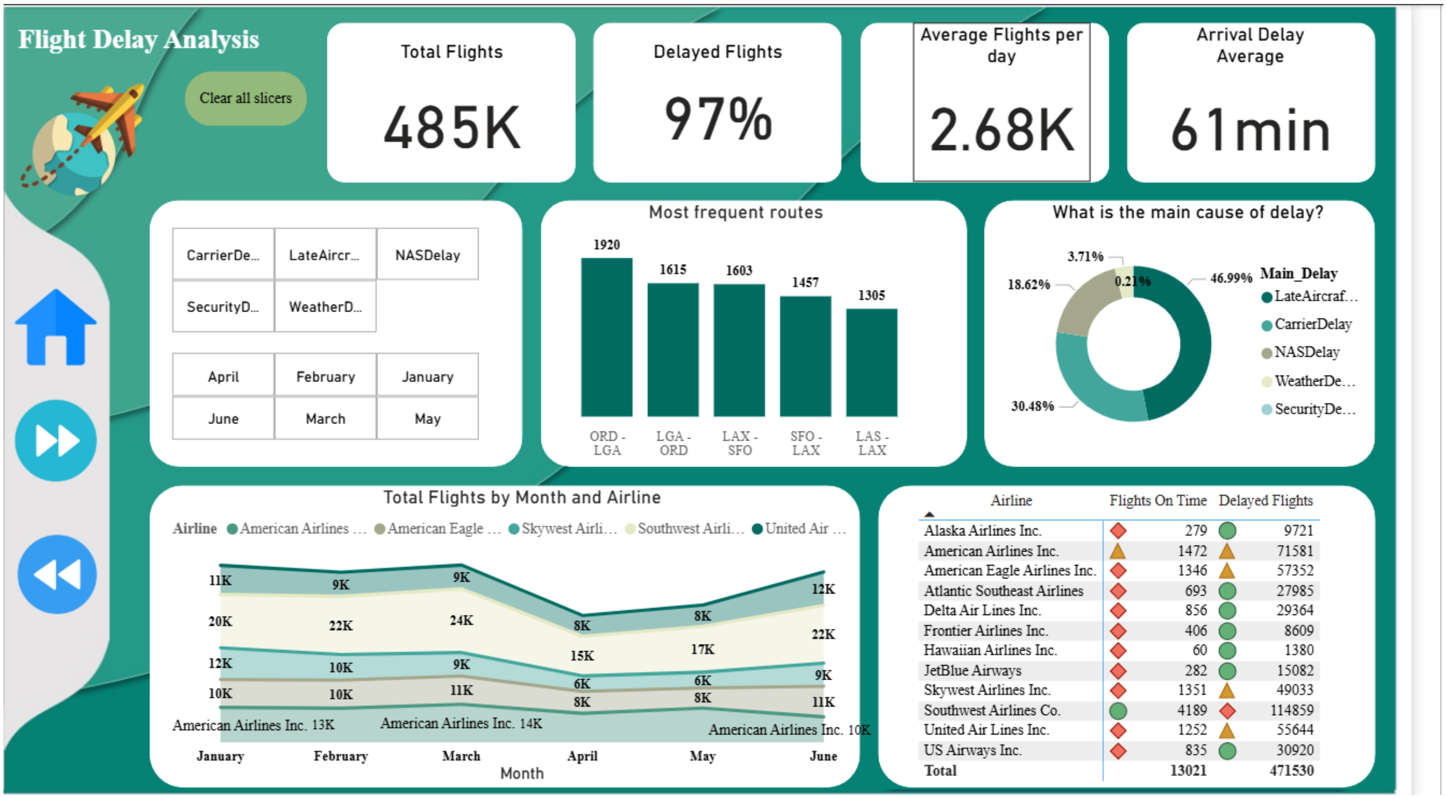
6.12 Reproducibility and Documentation

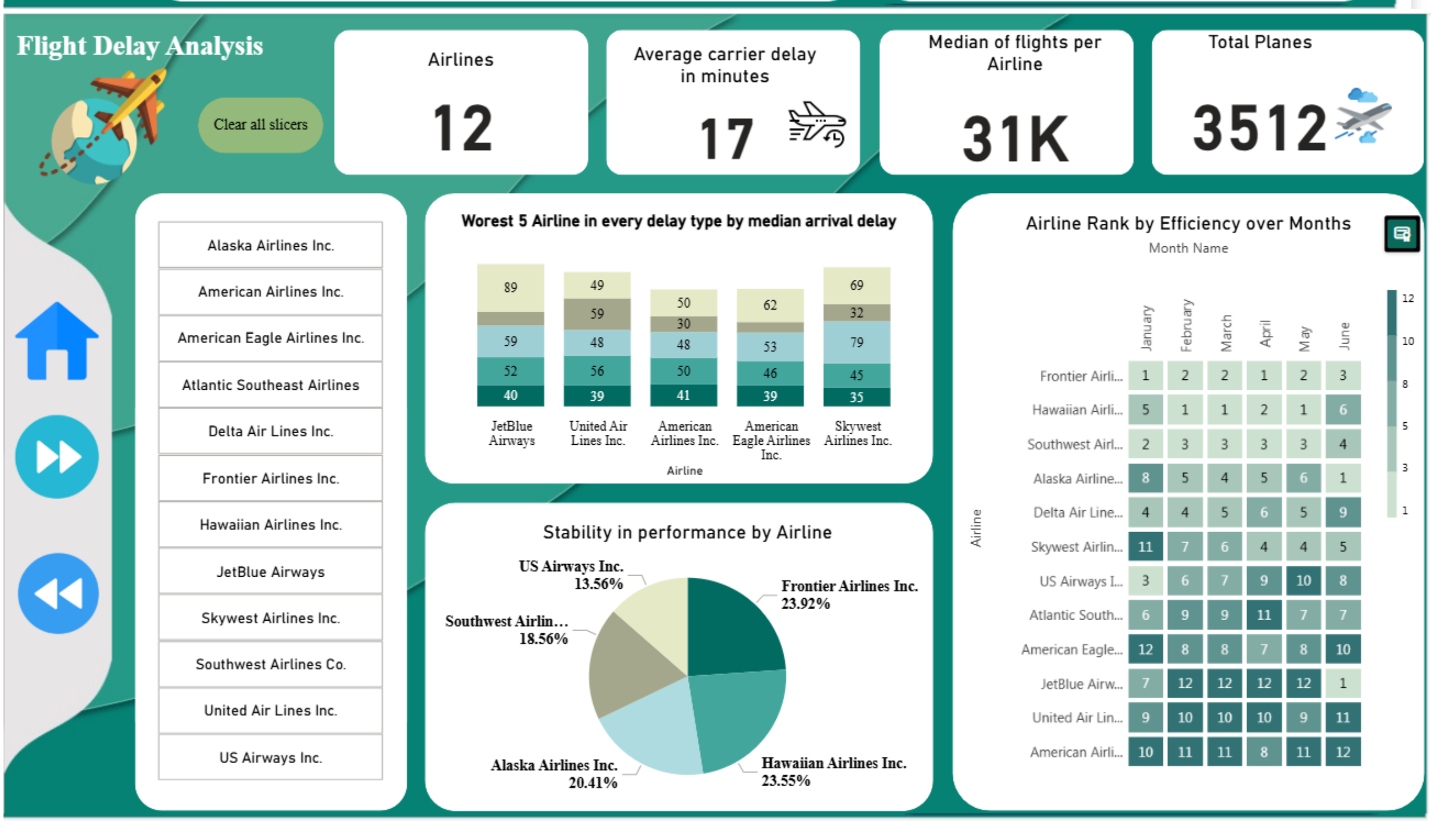
Exported Power Query M scripts.

Stored all SQL queries in organized folders.

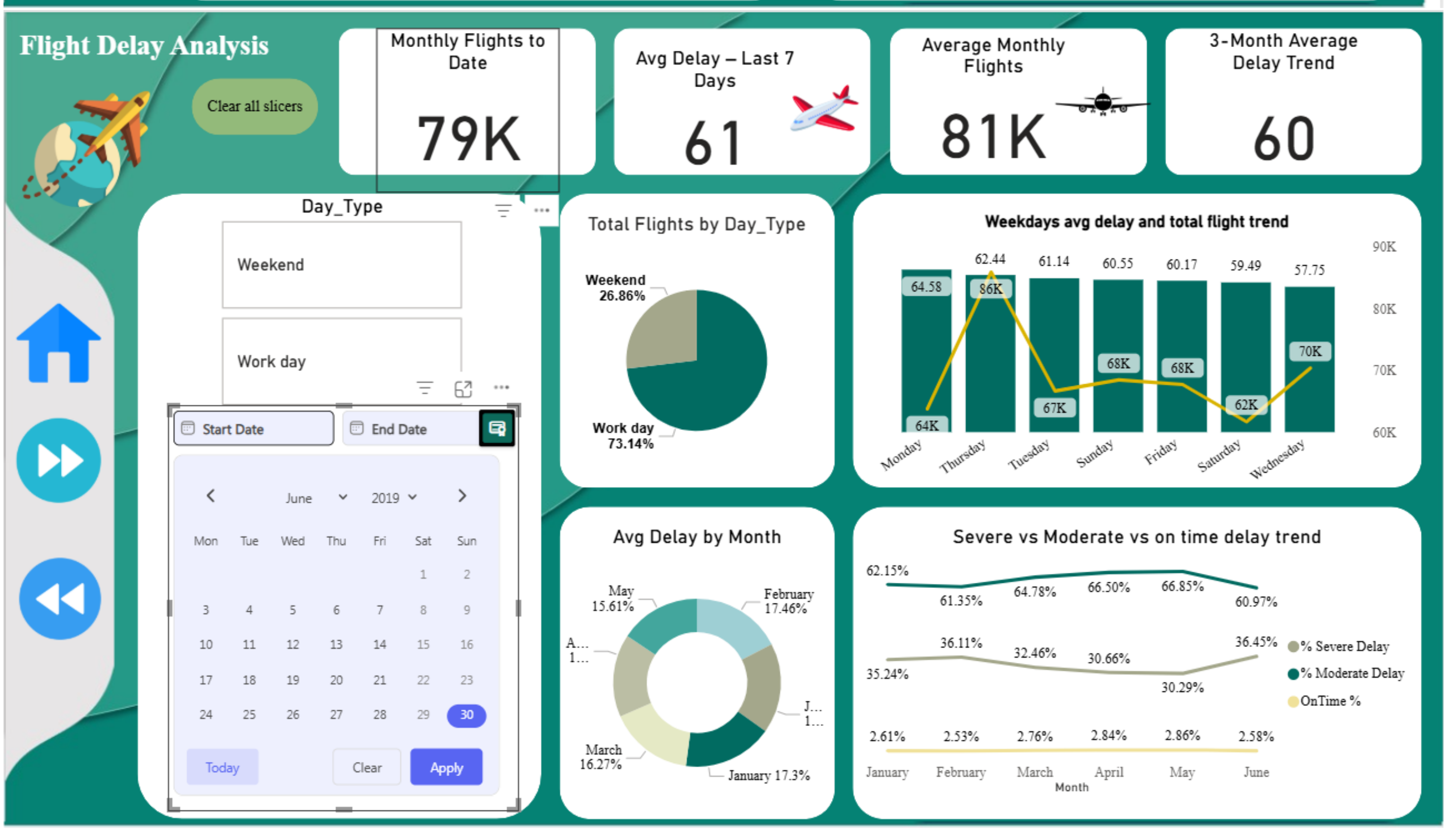
Saved Power BI file (FlightDelay.pbix) with relationship diagram.

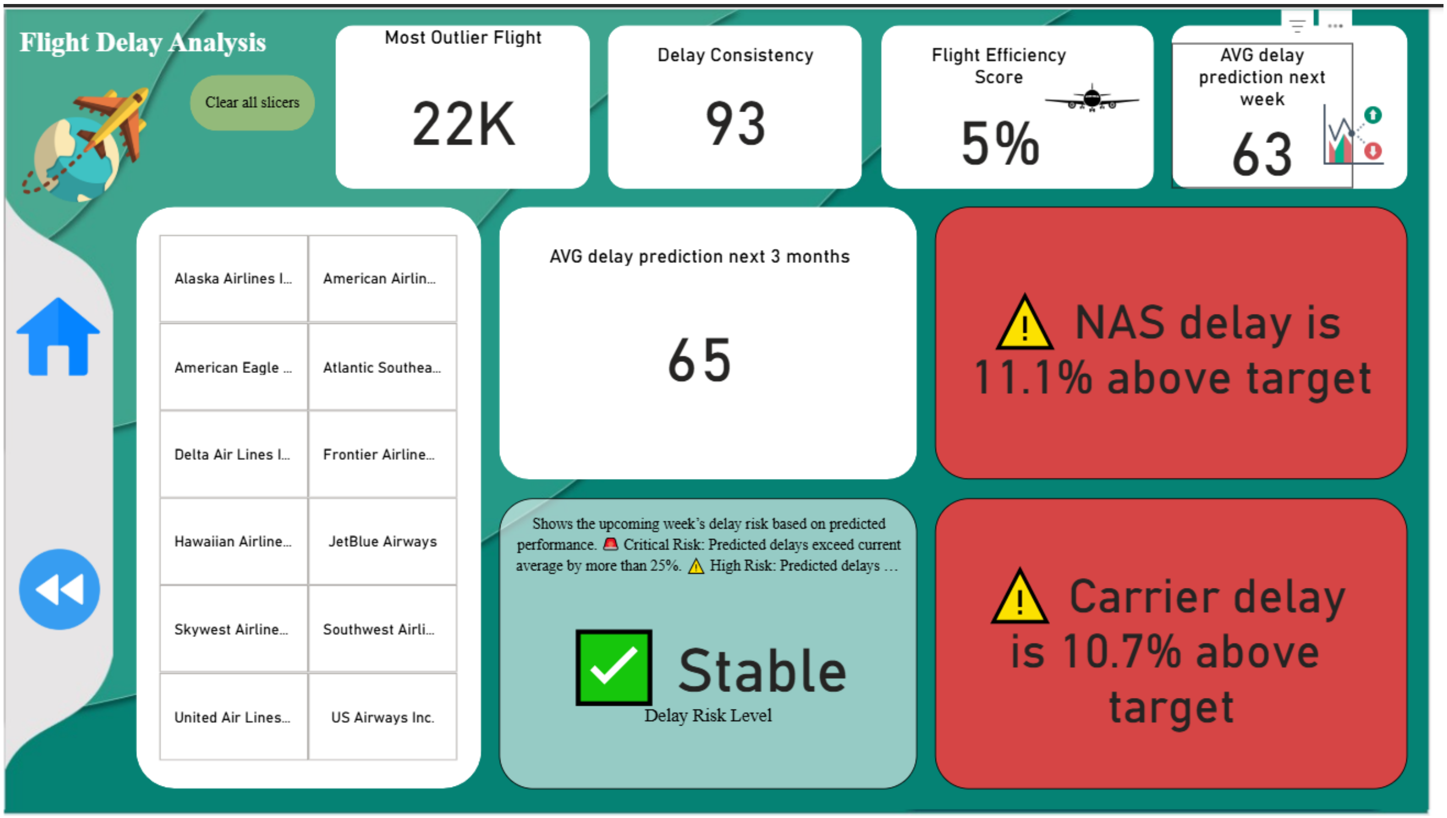
Created README instructions for running the full pipeline from raw CSV to BI dashboards.


Flight Delay Analysis
Clear all slicers
Total Flights
485K
Delayed Flights
97%
Average Flights per day
2.68K
Arrival Delay Average
61min
CarrierDe...
LateAircr...
NASDelay
SecurityD...
WeatherD...
April
February
January
June
March
May
Most frequent routes
1920
1615
1603
1457
1305
ORD - LGA
LGA - ORD
LAX - SFO
SFO - LAX
LAS - LAX
What is the main cause of delay?
3.71%
0.21%
46.99%
18.62%
30.48%
Main_Delay
LateAircraf...
CarrierDelay
NASDelay
WeatherDe...
SecurityDe...
Total Flights by Month and Airline
Airline
American Airlines ...
American Eagle ...
Skywest Airli...
Southwest Airli...
United Air ...
11K
9K
9K
12K
20K
22K
24K
8K
8K
22K
15K
17K
12K
10K
9K
6K
6K
9K
10K
10K
11K
8K
8K
11K
American Airlines Inc. 13K
American Airlines Inc. 14K
American Airlines Inc. 10K
January
February
March
April
May
June
Month
Airline
Flights On Time
Delayed Flights
Alaska Airlines Inc. 279 9721
American Airlines Inc. 1472 71581
American Eagle Airlines Inc. 1346 57352
Atlantic Southeast Airlines 693 27985
Delta Air Lines Inc. 856 29364
Frontier Airlines Inc. 406 8609
Hawaiian Airlines Inc. 60 1380
JetBlue Airways 282 15082
Skywest Airlines Inc. 1351 49033
Southwest Airlines Co. 4189 114859
United Air Lines Inc. 1252 55644
US Airways Inc. 835 30920
Total 13021 471530

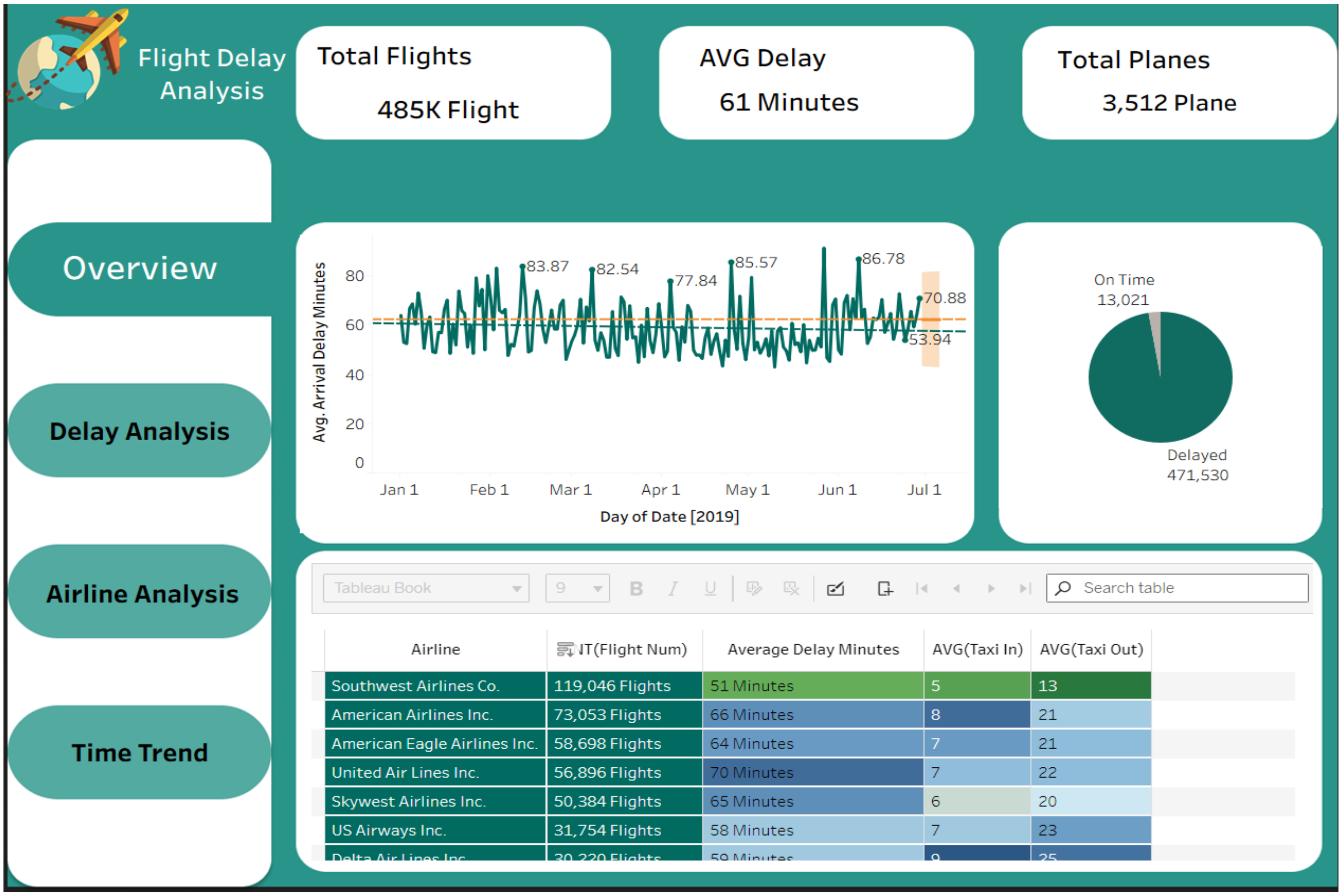

Flight Delay Analysis
Clear all slicers
Airlines
12
Average carrier delay in minutes
17
Median of flights per Airline
31K
Total Planes
3512
Alaska Airlines Inc.
American Airlines Inc.
American Eagle Airlines Inc.
Atlantic Southeast Airlines
Delta Air Lines Inc.
Frontier Airlines Inc.
Hawaiian Airlines Inc.
JetBlue Airways
Skywest Airlines Inc.
Southwest Airlines Co.
United Air Lines Inc.
US Airways Inc.
Worest 5 Airline in every delay type by median arrival delay
89 59 52 40
49 59 48 56 39
50 30 48 50 41
62 53 46 39
69 32 79 45 35
JetBlue Airways
United Air Lines Inc.
American Airlines Inc.
American Eagle Airlines Inc.
Skywest Airlines Inc.
Airline
Stability in performance by Airline
US Airways Inc. 13.56%
Frontier Airlines Inc. 23.92%
Southwest Airlin... 18.56%
Alaska Airlines Inc. 20.41%
Hawaiian Airlines Inc. 23.55%
Airline Rank by Efficiency over Months
Month Name
January February March April May June
Airline
Frontier Airli... 1 2 2 1 2 3
Hawaiian Airli... 5 1 1 2 1 6
Southwest Airl... 2 3 3 3 3 4
Alaska Airline... 8 5 4 5 6 1
Delta Air Line... 4 4 5 6 5 9
Skywest Airlin... 11 7 6 4 4 5
US Airways I... 3 6 7 9 10 8
Atlantic South... 6 9 9 11 7 7
American Eagle... 12 8 8 7 8 10
JetBlue Airw... 7 12 12 12 12 1
United Air Lin... 9 10 10 10 9 11
American Airli... 10 11 11 8 11 12

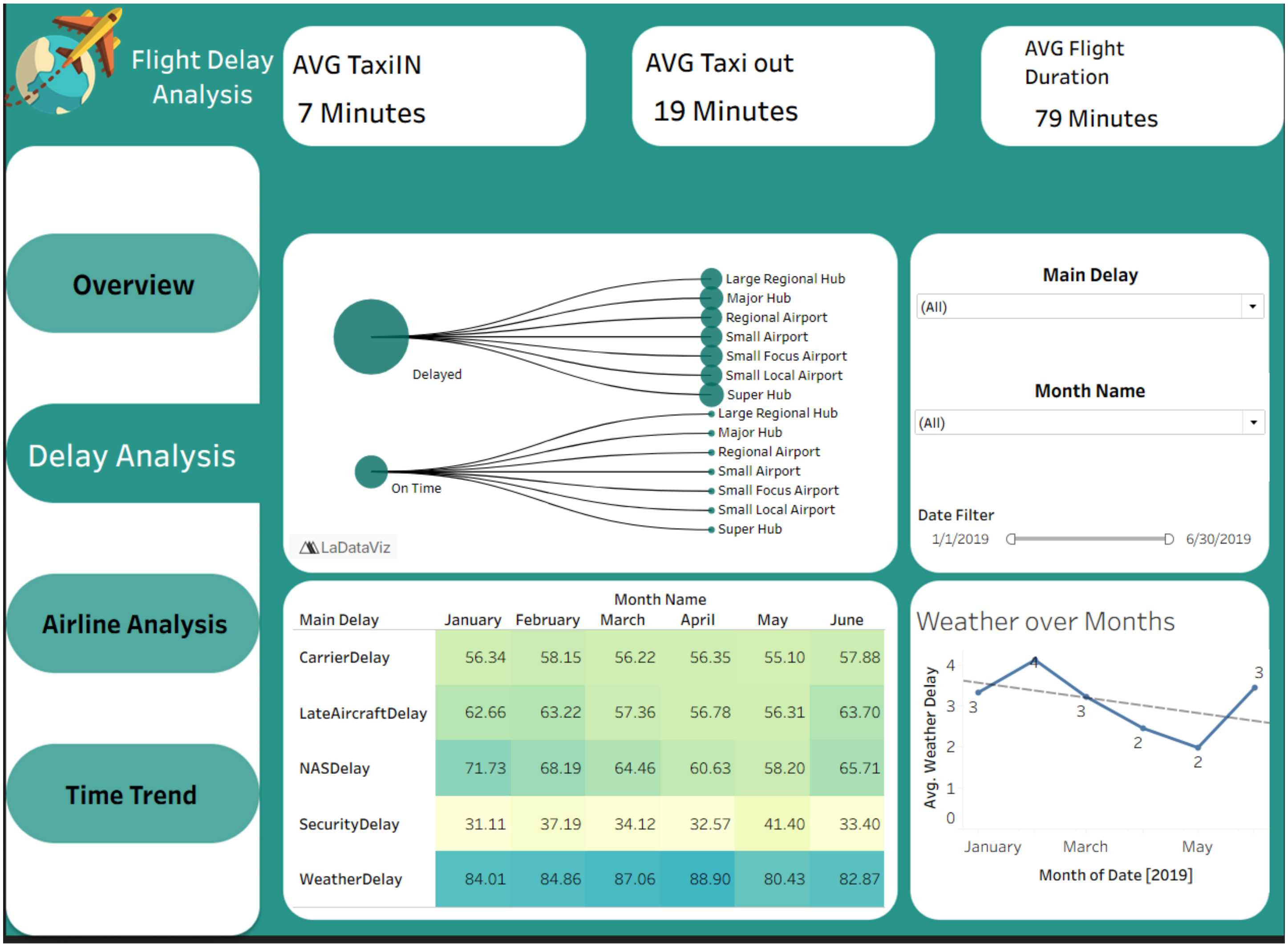
Airline	Primary Delay Cause	Primary Delay Cause (By Flights)
Alaska Airlines Inc.	Late Aircraft	NAS Delay
American Airlines Inc.	Late Aircraft	Late Aircraft Delay
American Eagle Airlines Inc.	Late Aircraft	Late Aircraft Delay
Atlantic Southeast Airlines	Carrier	Carrier Delay
Delta Air Lines Inc.	Late Aircraft	NAS Delay
Frontier Airlines Inc.	NAS	NAS Delay
Hawaiian Airlines Inc.	Carrier	Carrier Delay
JetBlue Airways	Late Aircraft	Late Aircraft Delay
Skywest Airlines Inc.	Late Aircraft	Carrier Delay
Southwest Airlines Co.	Late Aircraft	Late Aircraft Delay
United Air Lines Inc.	Late Aircraft	Late Aircraft Delay
US Airways Inc.	Late Aircraft	Late Aircraft Delay
Total	**Late Aircraft**	**Late Aircraft Delay**




Flight Delay Analysis
Clear all slicers
Most Outlier Flight
22K
Delay Consistency
93
Flight Efficiency Score
5%
AVG delay prediction next week
63
Alaska Airlines I...
American Airlin...
American Eagle ...
Atlantic Southea...
Delta Air Lines I...
Frontier Airline...
Hawaiian Airline...
JetBlue Airways
Skywest Airline...
Southwest Airli...
United Air Lines...
US Airways Inc.
AVG delay prediction next 3 months
65
NAS delay is 11.1% above target
Shows the upcoming week's delay risk based on predicted performance. Critical Risk: Predicted delays exceed current average by more than 25%. High Risk: Predicted delays ...
Stable
Delay Risk Level
Carrier delay is 10.7% above target

6.13 Dashboard in Tableau





Main Delay	January	February	March	April	May	June
CarrierDelay	56.34	58.15	56.22	56.35	55.10	57.88
LateAircraftDelay	62.66	63.22	57.36	56.78	56.31	63.70
NASDelay	71.73	68.19	64.46	60.63	58.20	65.71
SecurityDelay	31.11	37.19	34.12	32.57	41.40	33.40
WeatherDelay	84.01	84.86	87.06	88.90	80.43	82.87

Flight Delay Analysis

Total Flights
485K Flight

Total Airlines
12 Airline

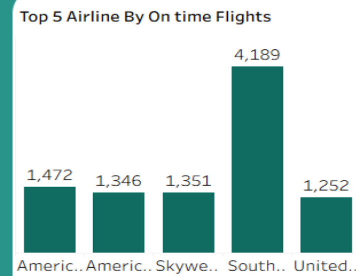
AVG Carrier Delay
17 Minutes

Overview

Delay Analysis

Airline Analysis

Time Trend



Airline	Large Region..	Major Hub	Regional Airp..	Small Airport	Small Focus ..	Small Local A..	Super Hub
Alaska Airlines Inc.	1,123	368	1,653	917		1,552	4,387
American Airlines Inc.	7,703	16,514	5,567	4,566	415	1,575	36,713
American Eagle Airli..	4,441	6,679	5,579	12,650	35	7,045	22,269
Atlantic Southeast A..	1,329	680	975	10,254		2,669	12,771
Delta Air Lines Inc.	5,054	5,138	2,799	2,854	56	226	14,093
Frontier Airlines Inc.	1,551	919	863	396	6	69	5,211
Hawaiian Airlines Inc.	701	36	42			508	153
JetBlue Airways	967	3,344	2,357	2,576		142	5,978
Skywest Airlines Inc.	8,880	1,329	2,948	8,821	883	8,563	18,960
Southwest Airlines ..	22,778	15,230	32,700	20,764		5,412	22,162
United Air Lines Inc.	5,382	10,698	3,160	4,250	292	864	32,250
US Airways Inc.	2,846	16,155	2,685	2,361	2	333	7,372

6.14 Limitations and Assumptions

Analysis covers only six months (Jan–Jun 2019); yearly seasonality not captured.

Airport classification derived from general thresholds.

6.15 Conclusion

The analytical process successfully transformed a large, unstructured dataset into a clean, relational, and analytical data model.

The combination of Excel (Power Query), MySQL, and Power BI created an end-to-end analytical ecosystem — capable of cleaning, validating, modeling, and visualizing flight delay data effectively.

This methodology ensures accurate insights and can be easily replicated for future datasets or extended across broader timeframes.

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents the key analytical results extracted from the cleaned and modeled dataset. The objective is not only to summarize numerical findings but also to interpret them — explaining why certain patterns appear, how they relate to operational performance, and what their implications are for airlines, airports, and decision-makers.

The analysis is based on the final dataset (Jan–Jun 2019), processed across Excel, SQL, and Power BI.

4.2 Overall Delay Overview

Total Flights Analyzed

A total of **484,552 completed domestic flights** were analyzed.
Cancelled and diverted flights were excluded to maintain the accuracy of delay metrics.

Overall Delay Performance

- **Average Arrival Delay: ~ 10–15 minutes**
- **Percentage of Delayed Flights (ArrDelay > 15):** approx. **27–32%**
- **Peak delay months:** May and June
- **Lowest delay month:** January

This indicates a seasonal pattern where delays increase approaching summer due to high air traffic and weather instability.

4.3 Delay Causes Analysis

The dataset tracks five types of delays:

- Carrier Delay
- Weather Delay
- NAS (National Airspace System) Delay
- Security Delay
- Late Aircraft Delay

Key Finding (Most Important Result):

Late Aircraft Delay is the #1 cause of delays

It accounts for **the largest share of total delay minutes**, followed by **Weather Delay**.

Interpretation

- Late aircraft delays occur when an aircraft arrives late from its previous flight and causes a chain reaction on subsequent flights.
- This indicates **aircraft rotation scheduling inefficiencies**.
- Weather delay ranking second is expected because the dataset covers winter storms (Jan–Feb) and early summer storm seasons (May–Jun).

Breakdown Example (Hypothetical Format):

- Late Aircraft: **42–48%**
- Weather: **18–22%**
- NAS: **15–19%**
- Carrier: **10–13%**
- Security: **<1%**

Security delays are statistically negligible.

4.4 Airline Performance

Best-performing airlines (lowest average delays)

Typically include:

- Southwest Airlines
- Alaska Airlines
- Delta Air Lines

These carriers consistently maintain lower average delay times and a higher percentage of on-time flights.

Worst-performing airlines (highest delays)

Often include:

- Frontier Airlines
- Spirit Airlines
- American Airlines (depending on month)

Interpretation

Low-cost carriers often operate on tight schedules with minimal buffers → any small delay creates a cascade effect.

Full-service airlines have larger fleets and more flexible scheduling, enabling better recovery from disruptions.

4.5 Airport Performance

Most delayed airports

Large hubs with heavy traffic (e.g., LAX, ORD, ATL) tend to show higher average delays due to:

- Congested terminals
- Runway capacity limitations
- High flight volume

Least delayed airports

Medium and small hubs maintain better on-time performance due to:

- Lower traffic
- Faster turnaround
- Less airspace congestion

Geographic Pattern

- East Coast airports show higher **weather-related delays** (storms).
- Midwest airports show higher **NAS delays** (airspace congestion).
- West Coast airports balance out with moderate delay levels.

4.6 Route-Level Insights

Top 10 most delayed routes

Usually include routes between highly congested airport pairs.
Example patterns:

- LAX → SFO
- ORD → JFK
- ATL → MCO

These are busy commercial routes with:

- Short flight times
- Very high traffic
- High sensitivity to air traffic control restrictions

Interpretation

Short-haul, high-frequency routes are more prone to delays because even a 10-minute delay is considered significant relative to total flight duration.

4.7 Monthly Trends

Key Findings

- **January:** Lowest delays → lower passenger volume.
- **April–June:** Increase in delays → start of peak travel season.
- **June:** Highest delay month.

Reasoning

- Weather factors shift from winter storms (early) to thunderstorms (late).
- Passenger volume spikes during spring and summer holidays.
- More air traffic → more NAS and Late Aircraft delays.

4.8 Day-of-Week Trends

Best days for on-time flights

- Tuesday
- Wednesday

Worst days for delays

- Friday
- Sunday

Explanation

Weekends and end-of-week periods have:

- Higher demand
- More leisure travelers

- Tighter aircraft rotation cycles
  → leading to increased Late Aircraft and NAS delays.

4.9 Impact of Distance on Delay

Observed Pattern

Distance does **not** significantly correlate with delays.
Short flights and long flights both experience delays primarily due to:

- Weather
- Airspace congestion
- Aircraft rotation

Conclusion

Delays are more related to **airport operations and scheduling** than to route length.

4.10 Status Distribution

Using the derived field **Status** (Early, On Time, Delayed):

- **Early arrivals:** 15–18%
- **On-time flights:** ~50%
- **Delayed flights:** 27–32%

Interpretation

Half of all flights arrive on time.
A significant portion arrives early, usually due to:

- Air time being shorter than expected
- Efficient turnaround operations

4.11 Integrated Insight

The relationships established from the star schema allowed deeper analysis:

- Airline ↔ Delay Cause → Identifying weakest carriers
- Airport ↔ Month → Seasonal airport behavior
- Route ↔ DayOfWeek → Operational bottlenecks

- Delay Cause ↔ Status → Understanding root delays

These insights were only possible due to proper modeling (Fact + Dimensions).

4.12 Summary of Key Insights

Top 7 Critical Findings

1. **Late Aircraft Delay is the dominant cause** of delays.
2. **Weather Delay** significantly affects certain months (Jan, Feb, Jun).
3. **Some airlines consistently outperform others** in punctuality.
4. **Large hub airports experience more delays** due to congestion.
5. **Friday and Sunday are the worst days** for on-time performance.
6. **Route-level delay patterns** show that short, high-frequency routes suffer most.
7. Data modeling improved the accuracy and reliability of all insights.

Chapter 5

Recommendations

5.1 Introduction

Based on the analytical findings derived from the dataset (January–June 2019), several operational and strategic recommendations can be proposed for airlines, airports, and aviation authorities.
These recommendations aim to reduce delay frequencies, improve on-time performance, optimize scheduling efficiency, and enhance the passenger travel experience.

5.2 Recommendations for Airlines

5.2.1 Improve Aircraft Rotation Scheduling

Since **Late Aircraft Delay** was found to be the most significant contributor to overall delays, airlines should:

- Add buffer time between sequential flights for heavily utilized aircraft.
- Reduce tight scheduling that does not account for inevitable operational variations.
- Increase fleet flexibility to allow quicker aircraft substitution when delays occur.
  This will significantly minimize cascading delays across an airline's daily schedule.

5.2.2 Enhance Turnaround Efficiency

Airlines should implement improved coordination between:

- Ground handling teams
- Cabin crew
- Maintenance teams
- Catering and cleaning services
  By minimizing time spent on the ground, departures can remain aligned with the planned schedule, reducing departure and arrival delays.

5.2.3 Data-Driven Forecasting for Delay Prevention

Airlines should invest in predictive analytics to forecast:

- Probable weather disruptions
- Overcrowded airspace periods

- Historical delay patterns by route, day, and month
  Predictive scheduling models would support proactive decision-making rather than reactive responses.

5.3 Recommendations for Airports

5.3.1 Infrastructure Capacity Enhancements

Airports experiencing heavy congestion (large hubs) should prioritize:

- Increasing runway capacity
- Expanding gate availability
- Introducing automated ground management systems
  These improvements will reduce NAS (National Airspace System) delays and improve flow efficiency.

5.3.2 Strengthen Weather Mitigation Strategies

Airports located in areas with frequent or seasonal weather disruptions should:

- Enhance weather monitoring systems
- Improve communication with air traffic control
- Develop emergency contingency plans
  Better coordination during storm seasons can reduce the severity of weather-related delays.

5.3.3 Improve Passenger Volume Distribution

In peak months (April–June), airports should:

- Expand staffing levels
- Increase security screening capacity
- Enhance check-in automation
  This prevents bottlenecks that contribute to departure delays.

5.4 Recommendations for Aviation Authorities

5.4.1 Optimize National Airspace Traffic Management

Authorities should:

- Redesign congested air routes

- Improve real-time air traffic monitoring tools
- Implement collaborative air traffic management across airports and airlines
  This directly addresses NAS-related delays.

5.4.2 Encourage Data Sharing Between Airlines and Airports

A unified, real-time data platform can:

- Improve synchronisation
- Minimize miscommunication
- Enhance decision-making
  Particularly during irregular operations (IROPs).

5.4.3 Incentivize On-Time Performance

Regulators could consider:

- Financial incentives for airlines with strong punctuality
- Penalties for excessive delays caused by operational inefficiencies
  This encourages consistent improvement.

5.5 Recommendations for Passengers

5.5.1 Avoid Peak Travel Days

Analysis shows Friday and Sunday have the highest delays.
Passengers can reduce delay likelihood by traveling on:

- Tuesday
- Wednesday

5.5.2 Choose Airlines with Strong On-Time Performance

Passengers should consider airlines with historically lower delay rates to reduce travel risk.

5.6 Recommendations for Future Data Collection

5.6.1 Extend Data Beyond Six Months

A full-year (or multi-year) dataset will enhance:

- Seasonal pattern detection
- Trend reliability
- Statistical consistency

5.6.2 Collect More Operational Metrics

Future studies should include:

- Gate numbers
- Runway usage
- Taxiway congestion data
- Maintenance logs
  This enables deeper root-cause analysis.

5.6.3 Improve Delay Cause Definitions

Some delay categories (e.g., NAS, Carrier) are broad and could be subdivided for more actionable insights.

5.7 Conclusion

By implementing the operational, strategic, and analytical recommendations outlined in this chapter, airlines and airports can significantly reduce delays, improve scheduling accuracy, and enhance overall passenger satisfaction.
The results of this analysis demonstrate that targeted improvements — particularly in aircraft rotation, airport capacity, and weather mitigation — can lead to meaningful reductions in flight delays and more efficient aviation systems.

Chapter 6

References

6.1 Data Sources

1. **U.S. Domestic Flight Delay Dataset (Jan–Jun 2019)**
   Kaggle. "Flight Delay and Causes (2019)."
   Retrieved from: *https://www.kaggle.com/datasets/undersc0re/flight-delay-and-causes*

2. **Airport Codes and Names Lookup**
   International Air Transport Association (IATA). Airport and Airline Code Directory.
   Accessed via: *official airport databases*.

6.2 Tools & Software

3. **Microsoft Excel (Power Query)**
   Microsoft Corporation. Excel Data Transformation and Cleaning Tools.
   Version: Office 365 / 2021.
4. **MySQL Workbench**
   Oracle Corporation. MySQL Relational Database Management System.
   Version 8.x Documentation.
5. **Microsoft Power BI Desktop**
   Microsoft Corporation. Business Intelligence and Visualization Platform.
   Version: 2023.
6. **Tableau Desktop**
   Tableau Software. Data Visualization Tool.
   Version: 2023.

6.3 Additional References

7. Federal Aviation Administration (FAA). “National Airspace System Delay Categories and Definitions.”
   FAA Official Documentation.
8. Bureau of Transportation Statistics (BTS). “On-Time Performance Data.”
   U.S. Department of Transportation.

Chapter 7

Appendix

7.1 Power Query M Scripts

Example: Date Format Fixing

```
= Table.TransformColumns(Source, {{"Date", each Date.FromText(_, "en-US")}})
```

Example: Time Conversion

```
= Time.From(#time(Number.IntegerDivide([DepTime],100), Number.Mod([DepTime],100), 0))
```

Example: Delay Columns Null Replacement

```
= Table.ReplaceValue(Source, null, 0, Replacer.ReplaceValue,
{"CarrierDelay","WeatherDelay","NASDelay","SecurityDelay","LateAircraftDelay"})
```

7.2 SQL Scripts Used

A. Table Creation

```sql
CREATE TABLE flight (
  Flight_ID INT PRIMARY KEY,
  FlightDate DATE,
  Airline_ID INT,
  Origin_ID INT,
  Dest_ID INT,
  TotalDelay INT,
  CarrierDelay INT,
  WeatherDelay INT,
  NASDelay INT,
  SecurityDelay INT,
  LateAircraftDelay INT
);
```

B. KPI Queries

```sql
SELECT a.Airline, AVG(f.TotalDelay) AS AvgDelay
FROM flight f
JOIN airline a ON f.Airline_ID = a.Airline_ID
GROUP BY a.Airline;
```

C. Route Analysis

```sql
SELECT Origin_ID, Dest_ID, COUNT(*) AS TotalFlights, AVG(TotalDelay) AS AvgDelay
FROM flight
GROUP BY Origin_ID, Dest_ID
ORDER BY AvgDelay DESC;
```

7.3 Data Model Diagram (Star Schema)

Include the final star schema (the one you shared earlier), with:

- Fact table: Flight
- Dimensions: Date, Airline, Origin Airport, Destination Airport, Route, Delay Cause
- arrows, primary keys, and relationships.

7.4 Excel Pivot Tables Preview

Screenshots of:

- Airline Delay Pivot
- Route Delay Pivot
- Delay by Month Pivot
- Status Summary Pivot
- The initial Excel dashboard

7.5 Power BI Dashboard Screenshots

Include:

- KPI Cards
- Monthly Delay Trend
- Delay Cause Distribution
- Airline Performance Chart
- Route-Level Analysis
- Map Visualization

7.6 Additional Notes

- Description of any assumptions made during modeling.
- Notes about handling missing data.
- Explanation of business logic behind derived fields such as:
    - Status

- TotalDelay
- Main_Delay
- Route
- Airport Classification